

1 You are given that $f(x) = x^4 - x^3 + x^2 + 9x - 10$.

(i) Show that $x = 1$ is a root of $f(x) = 0$ and hence express $f(x)$ as a product of a linear factor and a cubic factor.

[3]

(ii) Hence or otherwise find another root of $f(x) = 0$.

[2]

(iii) Factorise $f(x)$, showing that it has only two linear factors. Show also that $f(x) = 0$ has only two real roots.

[5]

2 You are given that $f(x) = x^2 + kx + c$.

Given also that $f(2) = 0$ and $f(-3) = 35$, find the values of the constants k and c .

[4]

3 You are given that $f(x) = (2x - 3)(x + 2)(x + 4)$.

(i) Sketch the graph of $y = f(x)$.

[3]

(ii) State the roots of $f(x - 2) = 0$.

[2]

(iii) You are also given that $g(x) = f(x) + 15$.

A Show that $g(x) = 2x^3 + 9x^2 - 2x - 9$.

[2]

B Show that $g(1) = 0$ and hence factorise $g(x)$ completely.

[5]

- 4 The table shows population data for a country.

Year	1969	1979	1989	1999	2009
Population in millions (p)	58.81	80.35	105.27	134.79	169.71

The data may be represented by an exponential model of growth. Using t as the number of years after 1960, a suitable model is $p = a \times 10^{kt}$.

- (i) Derive an equation for $\log_{10} p$ in terms of a , k and t .

[2]

- (ii) Complete the table and draw the graph of $\log_{10} p$ against t , drawing a line of best fit by eye.

[3]

- (iii) Use your line of best fit to express $\log_{10} p$ in terms of t and hence find p in terms of t .

[4]

- (iv) According to the model, what was the population in 1960?

[1]

- (v) According to the model, when will the population reach 200 million?

[3]

- 5 Fig. 9 shows a sketch of the curve $y = x^3 - 3x^2 - 22x + 24$ and the line $y = 6x + 24$.

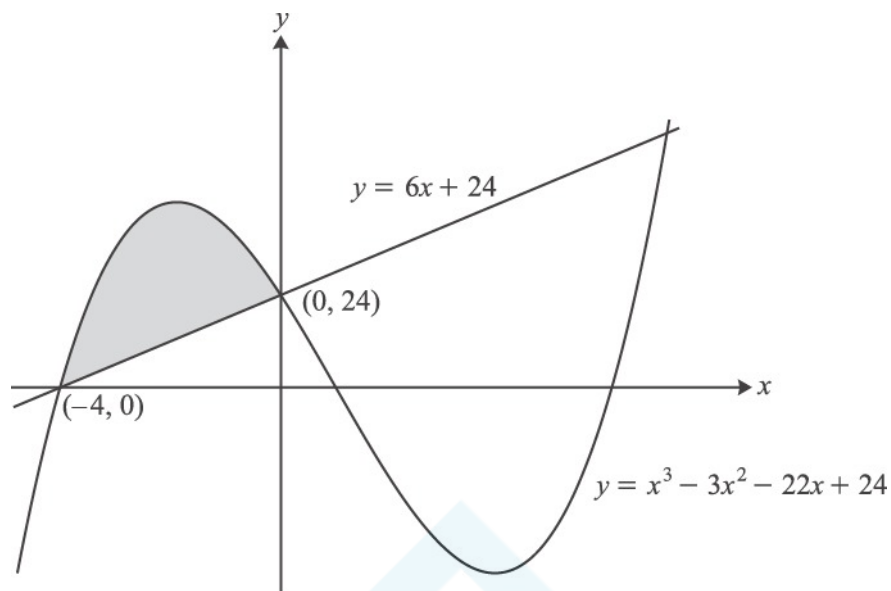


Fig. 9

- (i) Differentiate $y = x^3 - 3x^2 - 22x + 24$ and hence find the x -coordinates of the turning points of the curve. Give your answers to 2 decimal places. [4]
- (ii) You are given that the line and the curve intersect when $x = 0$ and when $x = -4$. Find algebraically the x -coordinate of the other point of intersection. [3]
- (iii) Use calculus to find the area of the region bounded by the curve and the line $y = 6x + 24$ for $-4 \leq x \leq 0$, shown shaded on Fig. 9. [4]

- 6 The thickness of a glacier has been measured every five years from 1960 to 2010. The table shows the reduction in thickness from its measurement in 1960.

Year	1965	1970	1975	1980	1985	1990	1995	2000	2005	2010
Number of years since 1960 (t)	5	10	15	20	25	30	35	40	45	50
Reduction in thickness since 1960 (h m)	0.7	1.0	1.7	2.3	3.6	4.7	6.0	8.2	12	15.9

An exponential model may be used for these data, assuming that the relationship between h and t is of the form $h = a \times 10^{bt}$, where a and b are constants to be determined.

- (i) Show that this relationship may be expressed in the form $\log_{10} h = mt + c$, stating the values of m and c in terms of a and b . [2]
- (ii) Complete the table of values in the answer book, giving your answers correct to 2 decimal places, and plot the graph of $\log_{10} h$ against t , drawing by eye a line of best fit. [4]
- (iii) Use your graph to find h in terms of t for this model. [4]
- (iv) Calculate by how much the glacier will reduce in thickness between 2010 and 2020, according to the model. [2]
- (v) Give one reason why this model will not be suitable in the long term. [1]

- 7 There are many different flu viruses. The numbers of flu viruses detected in the first few weeks of the 2012–2013 flu epidemic in the UK were as follows.

Week	1	2	3	4	5	6	7	8	9	10
Number of flu viruses	7	10	24	32	40	38	63	96	234	480

These data may be modelled by an equation of the form $y = a \times 10^{bt}$, where y is the number of flu viruses detected in week t of the epidemic, and a and b are constants to be determined.

- (i) Explain why this model leads to a straight-line graph of $\log_{10}y$ against t . State the gradient and intercept of this graph in terms of a and b . [3]
- (ii) Complete the values of $\log_{10}y$ in the table, draw the graph of $\log_{10}y$ against t , and draw by eye a line of best fit for the data.

Hence determine the values of a and b and the equation for y in terms of t for this model. [8]

During the decline of the epidemic, an appropriate model was

$$y = 921 \times 10^{-0.137w},$$

where y is the number of flu viruses detected in week w of the decline.

- (iii) Use this to find the number of viruses detected in week 4 of the decline. [1]

- 8 Express $1 < x < 3$ in the form $|x - a| < b$, where a and b are to be determined. [2]

The function $f(x)$ is defined by $f(x) = 1 - 2 \sin x$ for $-\frac{1}{2}\pi \leq x \leq \frac{1}{2}\pi$. Fig. 3 shows the curve $y = f(x)$.

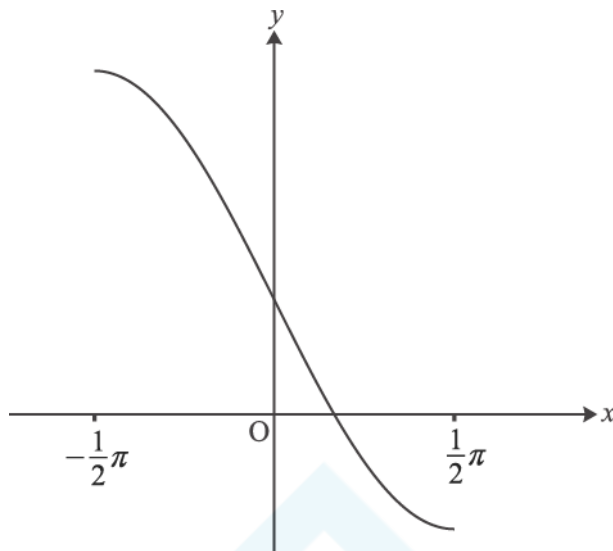


Fig. 3

- (i) Write down the range of the function $f(x)$.

[2]

- (ii) Find the inverse function $f^{-1}(x)$.

[3]

- (iii) Find $f'(0)$. Hence write down the gradient of $y = f^{-1}(x)$ at the point $(1, 0)$.

[3]

- 10 Fig. 1 shows the graphs of $y = |x|$ and $y = a|x + b|$, where a and b are constants. The intercepts of $y = a|x + b|$ with the x - and y -axes are $(-1, 0)$ and $(0, \frac{1}{2})$ respectively.

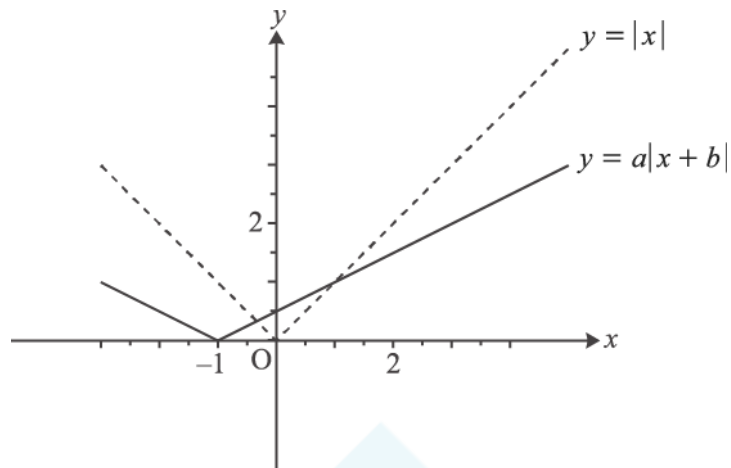


Fig. 1

- (i) Find a and b .

[2]

- (ii) Find the coordinates of the two points of intersection of the graphs.

[4]

11 Fig. 4 shows the curve $y = f(x)$, where

$$f(x) = a + \cos bx, 0 \leq x \leq 2\pi,$$

and a and b are positive constants. The curve has stationary points at $(0, 3)$ and $(2\pi, 1)$.

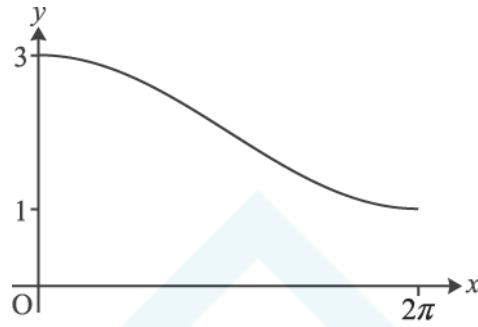


Fig. 4

(i) Find a and b .

[2]

(ii) Find $f^{-1}(x)$, and state its domain and range.

[5]

12 Solve the equation $|3 - 2x| = 4|x|$.

[4]

(i) The function $f(x)$ is defined by

$$f(x) = \frac{1-x}{1+x}, x \neq -1.$$

Show that $f(f(x)) = x$.

Hence write down $f^{-1}(x)$.

[3]

(ii) The function $g(x)$ is defined for all real x by

$$g(x) = \frac{1-x^2}{1+x^2}.$$

Prove that $g(x)$ is even. Interpret this result in terms of the graph of $y = g(x)$.

[3]

- 14 Fig. 9 shows the curve $y = f(x)$, where $f(x) = e^{2x} + k e^{-2x}$ and k is a constant greater than 1.

The curve crosses the y -axis at P and has a turning point Q .

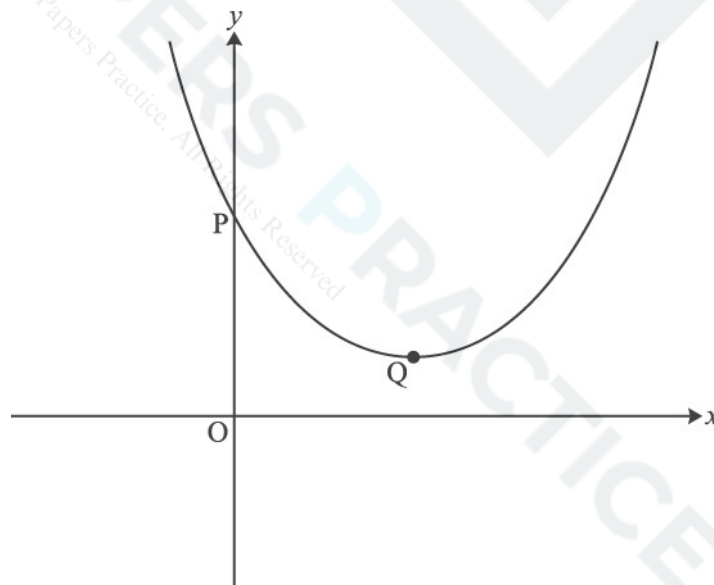


Fig. 9

(i) Find the y -coordinate of P in terms of k .

[1]

(ii) Show that the x -coordinate of Q is $\frac{1}{4}\ln k$, and find the y -coordinate in its simplest form.

[5]

(iii) Find, in terms of k , the area of the region enclosed by the curve, the x -axis, the y -axis and the line $x = \frac{1}{2}\ln k$.
Give your answer in the form $ak + b$.

[4]

The function $g(x)$ is defined by $g(x) = f(x + \frac{1}{4}\ln k)$.

(iv)

A Show that $g(x) = \sqrt{k} (e^{2x} + e^{-2x})$.

[3]

B Hence show that $g(x)$ is an even function.

[2]

C Deduce, with reasons, a geometrical property of the curve $y = f(x)$.

[3]

15 By sketching the graphs of $y = |2x + 1|$ and $y = -x$ on the same axes, show that the equation $|2x + 1| = -x$ has two roots. Find these roots.

[4]

16 The functions $f(x)$ and $g(x)$ are defined by $f(x) = \ln x$ and $g(x) = 2 + e^x$, for $x > 0$.

Find the exact value of x , given that $fg(x) = 2x$.

[5]

- 17 The speed of a 100 metre runner in ms^{-1} is measured electronically every 4 seconds.

EXAM PAPERS PRACTICE

The measurements are plotted as points on the speed-time graph in Fig. 6. The vertical dotted line is drawn through the runner's finishing time.

Fig. 6 also illustrates Model P in which the points are joined by straight lines.

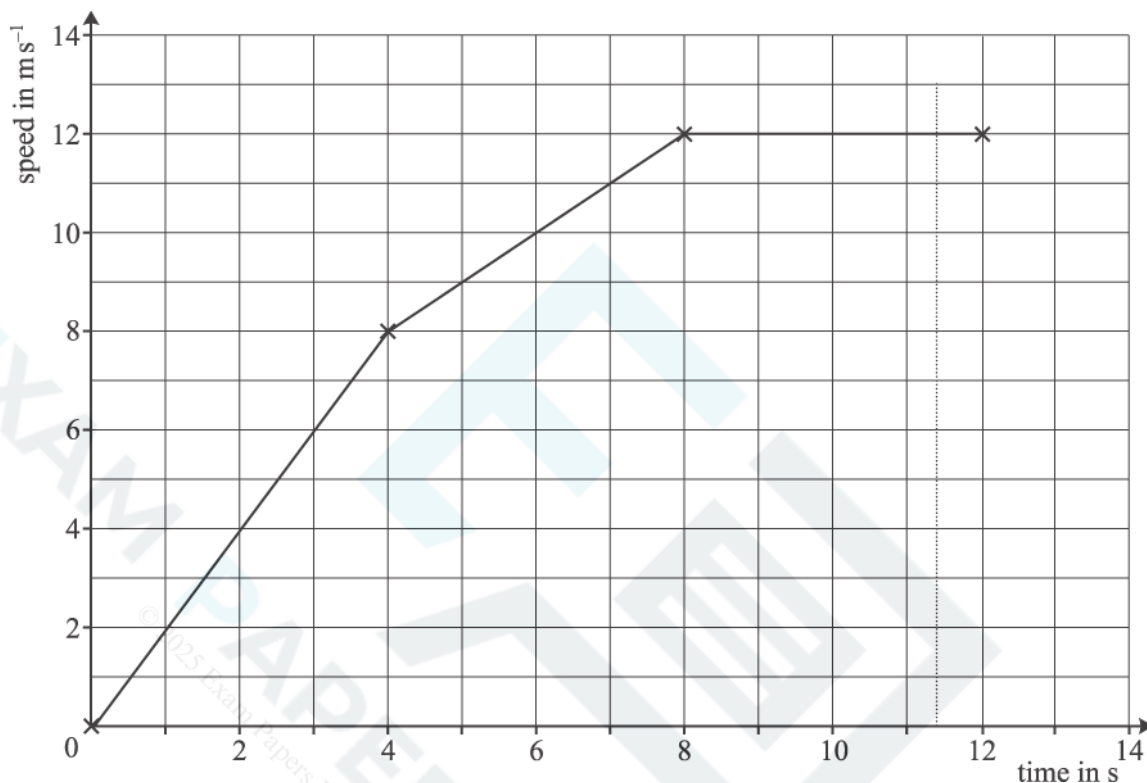


Fig. 6

(i) Use Model P to estimate

- A the distance the runner has gone at the end of 12 seconds,
- B how long the runner took to complete 100 m.

[6]

A mathematician proposes Model Q in which the runner's speed, $v \text{ ms}^{-1}$ at time $t \text{ s}$, is given by

(ii) Verify that Model Q gives the correct speed for $t = 8$.

[1]

(iii) Use Model Q to estimate the distance the runner has gone at the end of 12 seconds.

[4]

(iv) The runner was timed at 11.35 seconds for the 100 m.

Which model places the runner closer to the finishing line at this time?

[3]

(v) Find the greatest acceleration of the runner according to each model.

[4]

18 The function $f(x)$ is defined by $f(x) = x^3 - 4$ for $-1 \leq x \leq 2$.

For $f^{-1}(x)$, determine

- The domain
- The range.

[5]

19 In a chemical reaction, the mass m grams of a chemical at time t minutes is modelled by the differential equation

$$\frac{dm}{dt} = \frac{m}{t(1+2t)}.$$

At time 1 minute, the mass of the chemical is 1 gram.

(a) Solve the differential equation to show that $m = \frac{3t}{(1+2t)}.$

[8]

(b) Hence

(i) find the time when the mass is 1.25 grams,

[2]

(ii) show what happens to the mass of the chemical as t becomes large.

[2]

20 Solve the inequality $|2x - 1| \geq 4$.

[4]

21 The function $f(x)$ is defined by $f(x) = x^4 + x^3 - 2x^2 - 4x - 2$.

EXAM PAPERS PRACTICE

(a) Show that $x = -1$ is a root of $f(x) = 0$.

[1]

(b) Show that another root of $f(x) = 0$ lies between $x = 1$ and $x = 2$.

[2]

(c) Show that $f(x) = (x+1)g(x)$, where $g(x) = x^3 + ax + b$ and a and b are integers to be determined.

[3]

(d) Without further calculation, explain why $g(x) = 0$ has a root between $x = 1$ and $x = 2$.

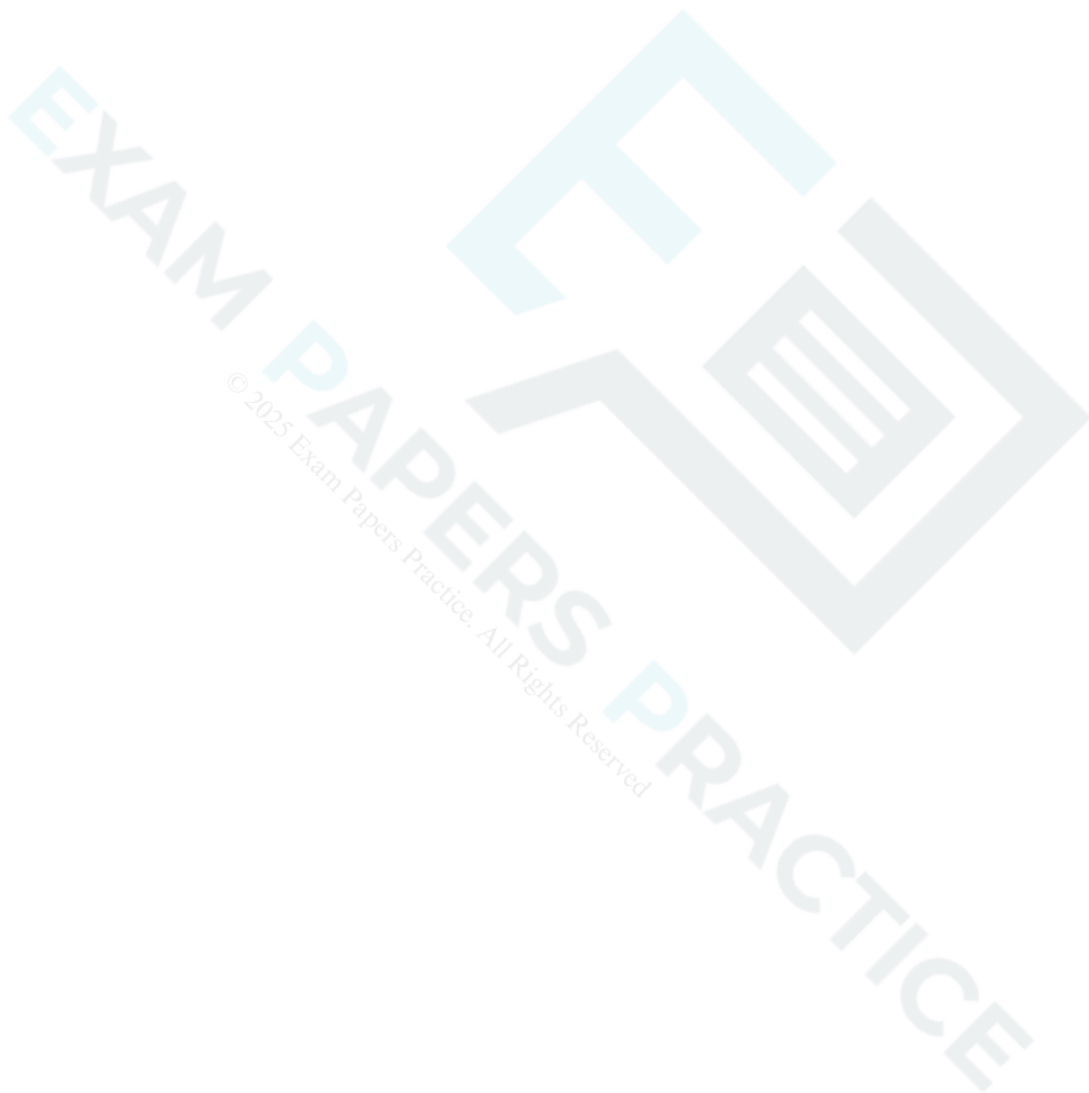
[1]

- (e) Use the Newton-Raphson formula to show that an iteration formula for finding roots of $g(x) = 0$ may be written

$$x_{n+1} = \frac{2x_n^3 + 2}{3x_n^2 - 2}$$

Determine the root of $g(x) = 0$ which lies between $x = 1$ and $x = 2$ correct to 4 significant figures.

[3]



22 You are given that $f(x) = 6x^3 - 25x^2 + 2x + 8$.

(a) Evaluate $f(4)$.

[1]

(b) In this question you must show detailed reasoning.

Express $f(x)$ as the product of three linear factors.

[4]

23

(a) Use the binomial expansion to show that $(1 - 2x)^{-\frac{1}{2}} \approx 1 + x + \frac{3}{2}x^2$ for sufficiently small values of x .

[2]

(b) For what values of x is the expansion valid?

[1]

(c) Find the expansion of $\sqrt{\frac{1+2x}{1-2x}}$ in ascending powers of x as far as the term in x^2 .

[3]

(d) Use $x = \frac{1}{20}$ in your answer to part (c) to find an approximate value for $\sqrt{11}$.

[2]



24 You are given that $gf(x) = |3x - 1|$, for $x \in \mathbb{R}$.



EXAM PAPERS PRACTICE

(a) Given that $f(x) = 3x - 1$, express $g(x)$ in terms of x .

[1]

(b) State the range of $gf(x)$.

[1]

(c) Solve the inequality $|3x - 1| > 1$.

[4]

EXAM PAPERS PRACTICE
© 2025 Exam Papers Practice. All Rights Reserved

25 In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

Fig. 2 shows the line $2x + y = 6$. The line crosses the x -axis at A and the y -axis at B.

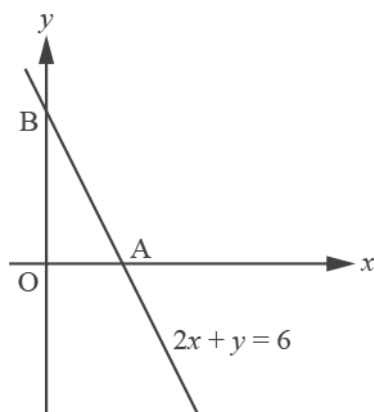


Fig. 2

Find the equation of the quadratic curve which touches the x -axis at A and passes through B.

[5]

A geometric series has first term $(b^2 - 13)$, common ratio $\frac{1}{b}$ and sum to infinity -6 .

Find the possible values of the common ratio.

[9]



27 The function $f(x)$ is defined by $f(x) = |x|$, for $-1 \leq x \leq 1$.

EXAM PAPERS PRACTICE

Sketch the graph of $y = g(x)$, where $g(x) = 2 - 2f(x)$.

[3]



28 Functions $f(x)$ and $g(x)$, each defined for $-1 < x < 1$, are given by $f(x) = \ln(1 - x)$ and $g(x) = x^2$.

EXAM PAPERS PRACTICE

(i) Find $f^{-1}(x)$ and state its domain and range.

[4]

(ii) Show that $f(x) + f(-x) = fg(x)$.

[3]

29 (a) Show that $(x - 3)$ is a factor of $f(x) = 6x^3 - 17x^2 - 5x + 6$.

[1]

EXAM PAPERS PRACTICE

(b) Show that $f(x)$ can be written as the product of 3 linear factors.

[4]



30 The function $f(x)$ is defined by $f(x) = x^4 - 3x^3 + 3x^2$ for $x \in \mathbb{R}$.

EXAM PAPERS PRACTICE

(a) Show that the only stationary point on the curve $y = f(x)$ is a minimum point at the origin.

[7]



(b) Explain why $f(x)$ does not have an inverse function.

[1]

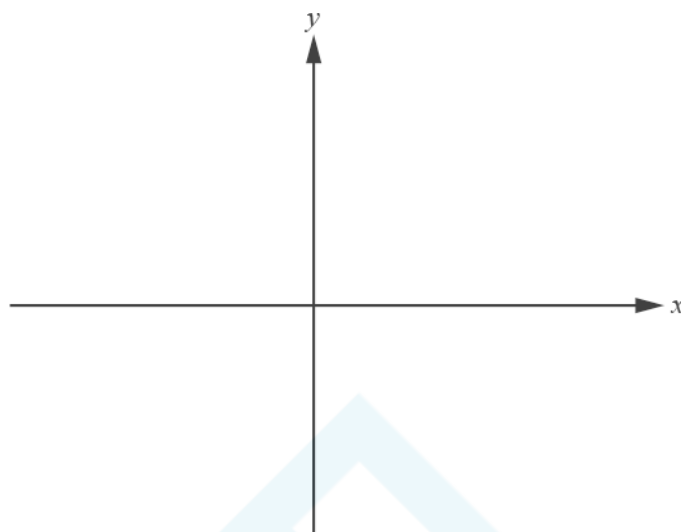
The function $g(x)$ is defined by $g(x) = x^4 - 3x^3 + 3x^2$ for $0 \leq x \leq 2$.

(c) State the domain and range of the inverse function $g^{-1}(x)$.

[2]

(a) On the same axes, sketch the graphs of $y = x$ and $y = |2x - 1|$.

[2]



EXAM PAPERS PRACTICE
© 2025 Exam Papers Practice. All Rights Reserved

(b) In this question you must show detailed reasoning.

Solve the inequality $|2x - 1| > x$.

[4]



32 Show that $(x - 2)$ is a factor $3x^3 - 8x^2 + 3x + 2$.



EXAM PAPERS PRACTICE

[3]

33 Solve the equation $|2x + 1| < 5$.

[3]

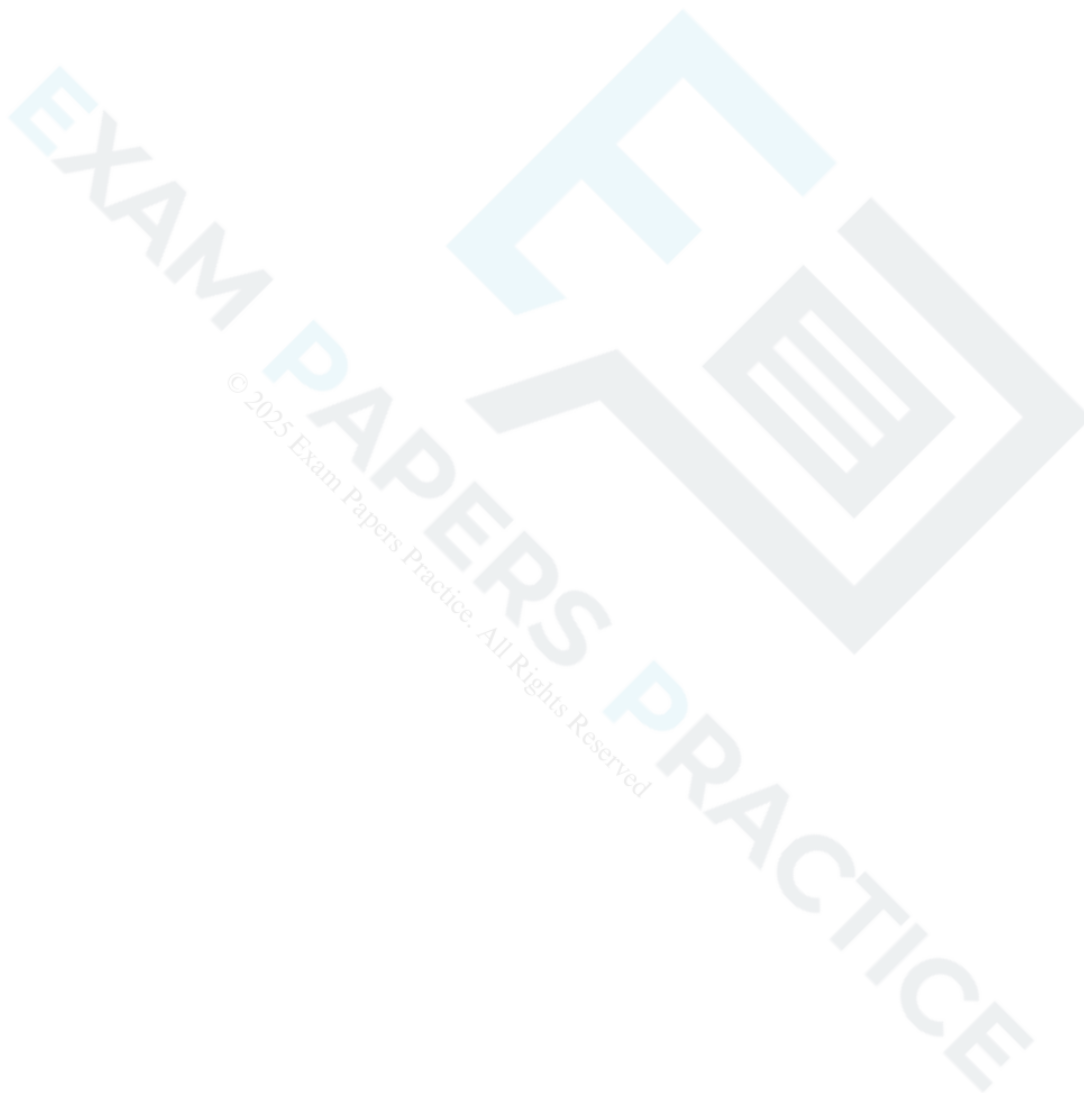
- 34 In the first year of a course, an A-level student, Aaishah, has a mathematics test each week. The night before each test she revises for t hours. Over the course of the year she realises that her percentage mark for a test, p , may be modelled by the following formula, where A , B and C are constants.

$$p = A - B(t - C)^2$$

- Aaishah finds that, however much she revises, her maximum mark is achieved when she does 2 hours revision. This maximum mark is 62.
- Aaishah had a mark of 22 when she didn't spend any time revising.

(a) Find the values of A , B and C .

[3]



- (b) According to the model, if Aaishah revises for 45 minutes on the night before the test, what mark will she achieve? [2]

- (c) What is the maximum amount of time that Aaishah could have spent revising for the model to work? [2]



In an attempt to improve her marks Aaishah now works through problems for a total of t hours over the three nights before the test. After taking a number of tests, she proposes the following new formula for p .

$$p = 22 + 68(1 - e^{-0.8t})$$

For the next three tests she recorded the data in Fig. 16.

t	1	3	5
p	59	84	89

Fig. 16

- (d) Verify that the data is consistent with the new formula.

[2]

- (e) Aaishah's tutor advises her to spend a minimum of twelve hours working through problems in future. Determine whether or not this is good advice.

[2]

35 (a) Sketch the graphs of $y = 4 \cos x$ and $y = 2 \sin x$ for $0^\circ \leq x \leq 180^\circ$ on the same axes.

[2]

(b) Find the exact coordinates of the point of intersection of these graphs, giving your answer in the form $(\arctan a, k\sqrt{b})$, where a and b are integers and k is rational.

[4]

(c) A student argues that without the condition $0^\circ \leq x \leq 180^\circ$ all the points of intersection of the graphs would occur at intervals of 360° because both $\sin x$ and $\cos x$ are periodic functions with this period. Comment on the validity of the student's argument.

[1]

36 In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

You are given that $f(x) = 4x^3 - 3x + 1$.

(a) Use the factor theorem to show that $(x + 1)$ is a factor of $f(x)$.

[2]

(b) Solve the equation $f(x) = 0$.

[3]

© 2025 Exam Papers Practice. All Rights Reserved

- (a) Show that $x - 3$ is a factor of $4x^3 - 12x^2 - x + 3$.

[1]

- (b) Fig. 4 shows the curve $y = 4x^3 - 12x^2 - x + 3$. Find the coordinates of the points where it crosses the x -axis.

[4]

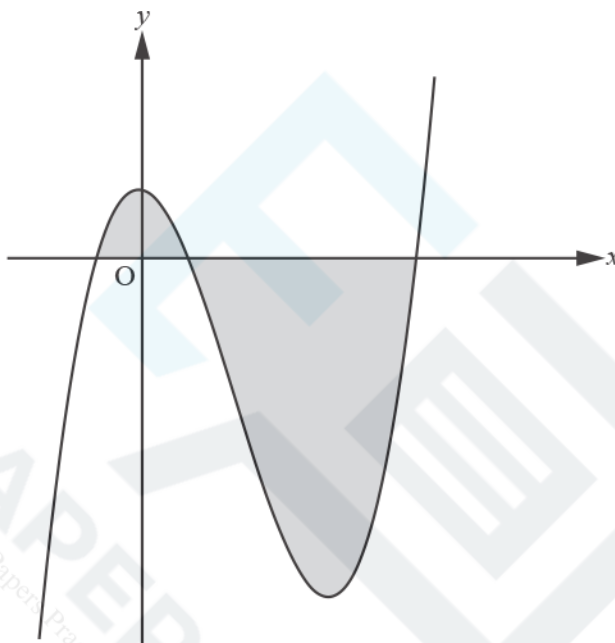
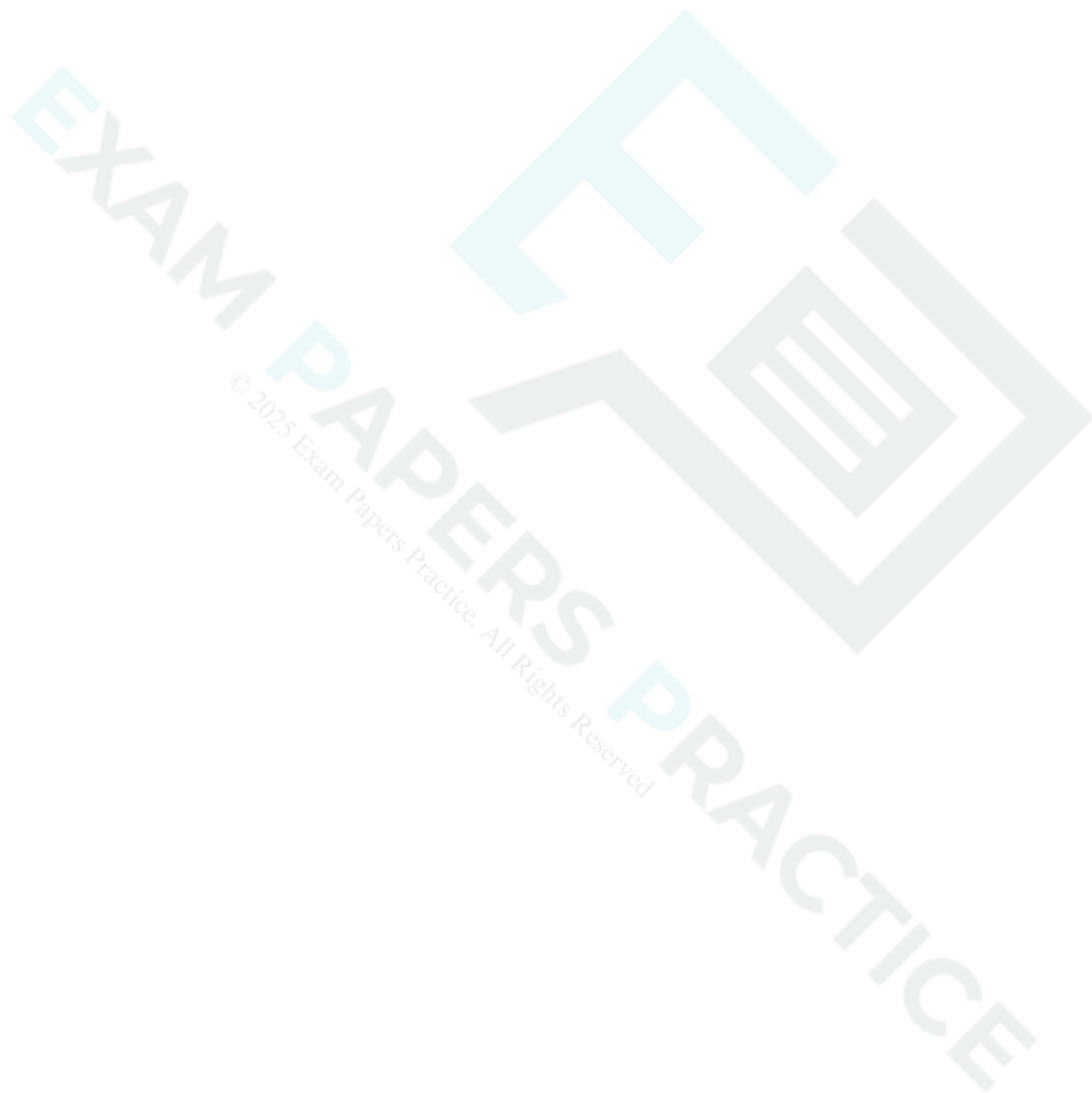


Fig. 4

- (c) The two regions bounded by the curve $y = 4x^3 - 12x^2 - x + 3$ and the x -axis are shaded in Fig. 4. Determine the total area of the shaded regions.

[5]



38 (see insert for practice3 H640/03) This question is about a straight line demand curve with equation $P = mQ + c$, where $m < 0$ and $c > 0$.

(a) Find an expression for the revenue in terms of Q , m and c .

[1]

(b) Hence show that, in this case, the maximum revenue occurs where the PED is -1 , as stated on line 35.

[5]

39 You are given that

$$f(x) = x^4 - x, \quad x > 1.$$

The graphs of $y = f(x)$, $y = x$ and $y = f^{-1}(x)$ are shown in Fig. 9.

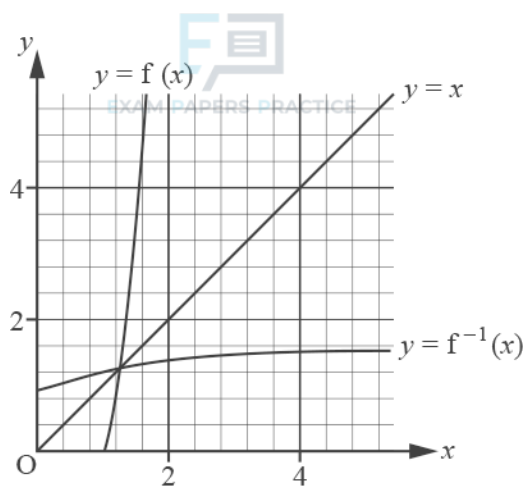


Fig. 9

(a) Show that the inverse function, $f^{-1}(x)$, passes through the point (14, 2).

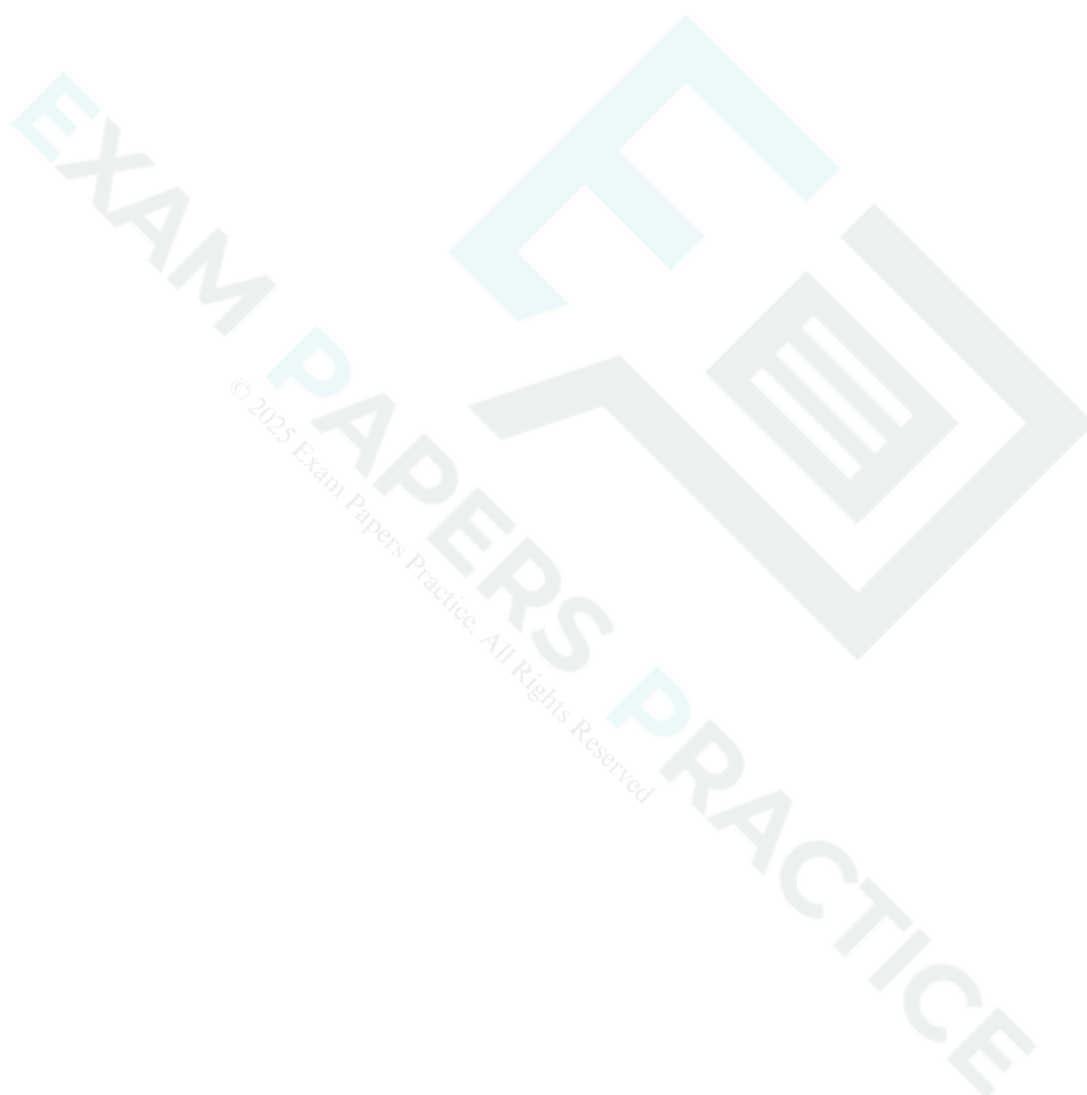
[1]

(b) Find the gradient of $f^{-1}(x)$ at the point (14, 2).

[3]

40 Solve the equation $|4x - 5| = 3$.

[3]





A function is defined by $f(x) = \frac{2x-3}{x-2}$ for $x \in \mathbb{R}$, $x \neq 2$.

(a) Show that $ff(x) = x$.

[3]

(b) Hence write down $f^{-1}(x)$.

[1]



- (c) The graph of $y = f(x)$ is translated by $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$. Find the equation of the translated graph, giving your expression for y in the form of a simplified algebraic fraction. [3]

- 42 Find the quotient and the remainder when $x^3 - 2x + 3$ is divided by $x + 2$. [3]

- 43 Write down the roots of the equation $(x + 2)(x^2 - 25) = 0$. [2]

44 The population of a small country is modelled using the formula $P = 5 \times 1.02^n$ where P is the population in millions and n is the number of years after the start of the year 2000.

(a) According to the model, what is the population of the country at the start of the year 2000? [1]

(b) Explain fully what the model implies about how the population changes over time. [2]

(c) In this question you must show detailed reasoning.

According to the model, in what year will the population reach 10 million? [3]

(d) Show that, according to the model, the graph of $\log_{10} P$ against n will be a straight line. [2]



The function $f(x) = \frac{1}{x}$ is defined for the domain $\{x: x \in \mathbb{R}, x \neq 0\}$. The function $g(x)$ is defined for all real x . The

composite function $fg(x)$ is defined by $fg(x) = \frac{1}{3x-6}$.

(a) Write down an expression for $g(x)$. [1]

(b) What is the domain of $fg(x)$? [1]

(c) Describe a sequence of two transformations that would map the graph of $y = f(x)$ onto the graph of $y = fg(x)$. [4]

EXAM PAPERS PRACTICE
© 2025 Exam Papers Practice. All Rights Reserved

46(a) In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

Nigel is asked to determine whether $(x + 7)$ is a factor of $x^3 - 37x + 84$. He substitutes $x = 7$ and calculates $7^3 - 37 \times 7 + 84$. This comes to 168, so Nigel concludes that $(x + 7)$ is not a factor.

Nigel's conclusion is wrong.

- Explain why Nigel's argument is not valid.
- Show that $(x + 7)$ is a factor of $x^3 - 37x + 84$.

[2]

- (b) Sketch the graph of $y = x^3 - 37x + 84$, indicating the coordinates of the points at which the curve crosses the coordinate axes.

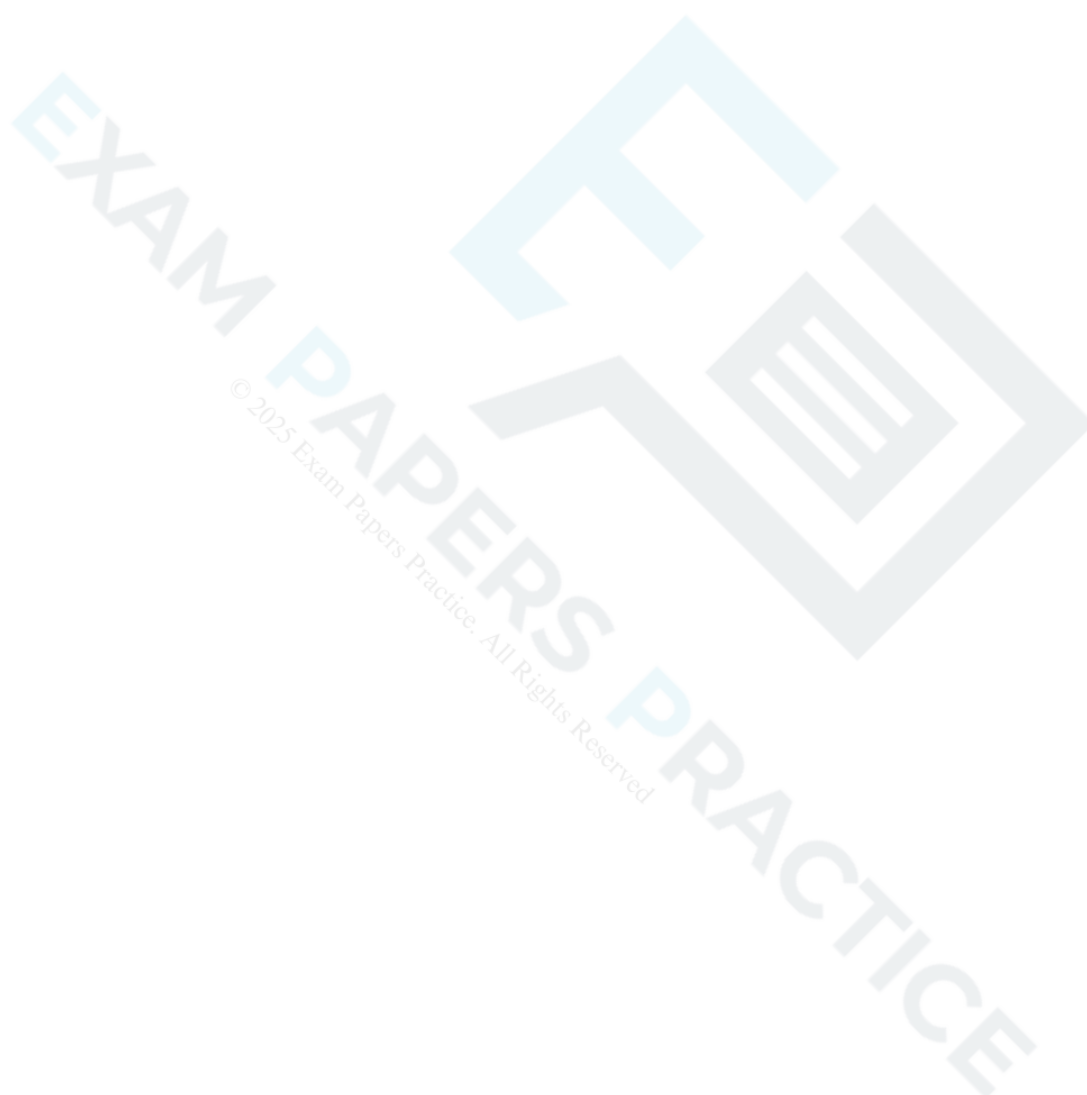
[5]

© 2025 Exam Papers Practice. All Rights Reserved

(c)

The graph in the above part is translated by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Find the equation of the translated graph, giving your answer in the form $y = x^3 + ax^2 + bx + c$ where a , b and c are integers.

[4]



47(a) You are given that

$$f(x) = 2x + 3 \quad \text{for } x < 0$$

and

$$g(x) = x^2 - 2x + 1 \quad \text{for } x > 1.$$

Find $gf(x)$, stating the domain.

[3]

(b) State the range of $gf(x)$.

[1]

(c) Find $(gf)^{-1}(x)$.

[5]

- (i) Show that the cross-sectional area in Fig. C3.2 is $\pi x(2r-x)$.

[2]

(ii)

Hence show that the cross-sectional area is $\frac{\pi r^2}{h^2}(h^2 - y^2)$, as given in line 37.

[2]

(b)

Verify that the formula $\frac{\pi r^2}{h^2}(h^2 - y^2)$ for the cross-sectional area is also valid for

- (i) Fig. C3.1,

[1]

- (ii) Fig. C3.3.

[1]

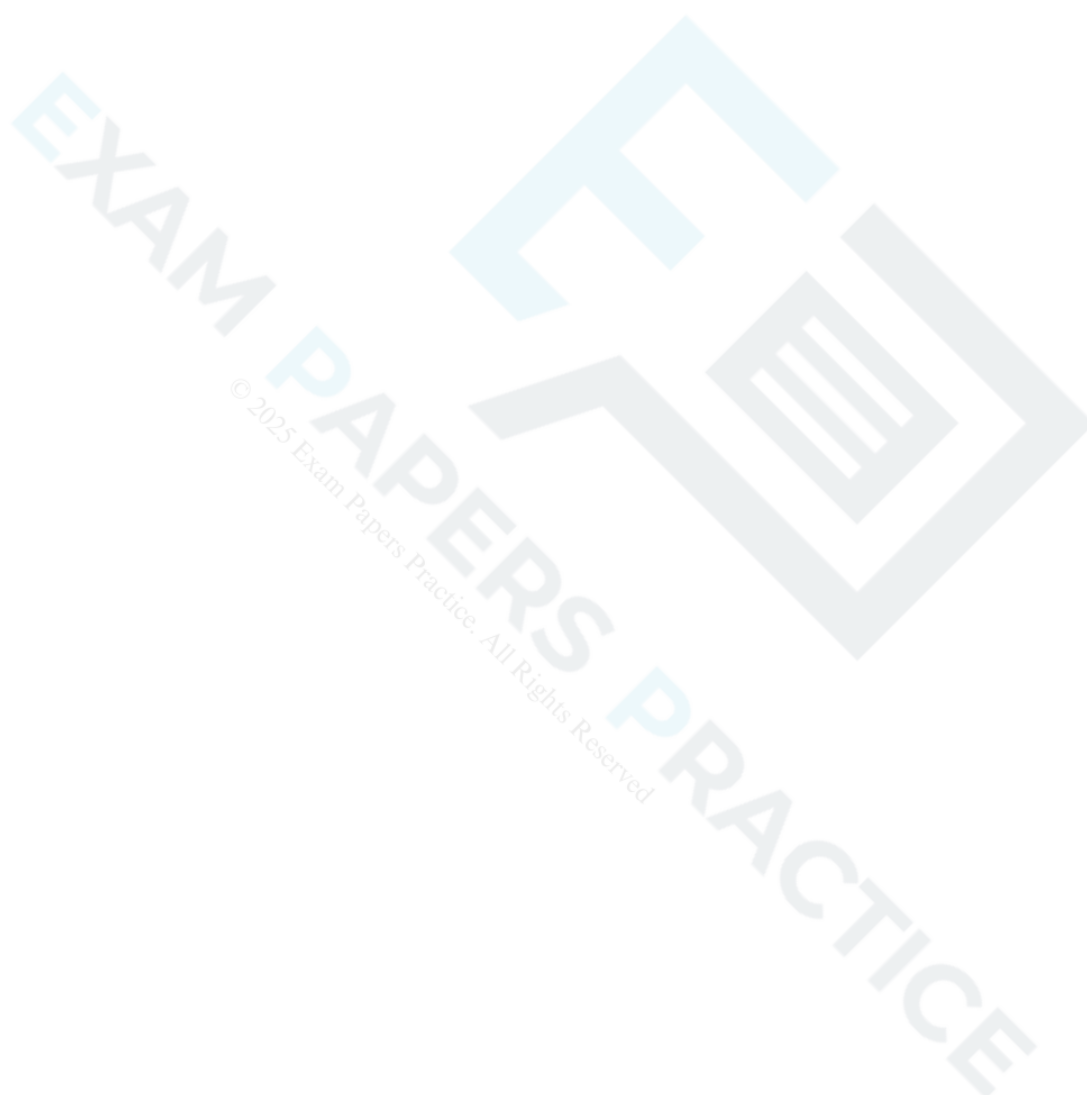
49 To answer this question, please refer to the Insert for H640-03 June 2019.

EXAM PAPERS PRACTICE

A typical tube of toothpaste measures 5.4 cm across the straight edge at the top and is 12 cm high. It contains 75 ml of toothpaste so it needs to have an internal volume of 75cm^3 .

Comment on the accuracy of the formula $V = \frac{2}{3}\pi r^2 h$, as given in line 41, for the volume in this case.

[3]



50(a) A uniform ruler AB has mass 28 g and length 30 cm. As shown in Fig. 6, the ruler is placed on a horizontal table so that it overhangs a point C at the edge of the table by 25 cm.

A downward force of F N is applied at A. This force just holds the ruler in equilibrium so that the contact force between the table and the ruler acts through C.



Fig. 6

Complete the force diagram below, labelling the forces and all relevant distances.

[2]



(b) Calculate the value of F .

[2]

51(a) In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

The function $f(x)$ is defined by $f(x) = x^3 + x^2 - 8x - 12$ for all values of x .

Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$.

[2]

(b) Solve the equation $f(x) = 0$.

[4]

EXAM PAPERS PRACTICE
© 2025 Exam Papers Practice. All Rights Reserved

52(a) In this question, the x , and y directions are horizontal and vertically upwards respectively.

EXAM PAPERS PRACTICE

A particle of mass 1.5 kg is in equilibrium under the action of its weight and forces $\mathbf{F}_1 = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \text{ N}$ and $\mathbf{F}_2$.

Find the force $\mathbf{F}_2$.

[3]

(b)

The force $\mathbf{F}_2$ is changed to $\begin{pmatrix} 2 \\ 20 \end{pmatrix} \text{ N}$.

Find the acceleration of the particle.

[3]

53(a) An astronaut on the surface of the moon drops a ball from a point 2 m above the surface.

EXAM PAPERS PRACTICE

Without any calculations, explain why a standard model using $g = 9.8 \text{ ms}^{-2}$ will not be appropriate to model the fall of the ball.

[1]

(b) The ball takes 1.6 s to hit the surface.

Find the acceleration of the ball which best models its motion. Give your answer correct to 2 significant figures.

[2]

(c) Use this value to predict the maximum height of the ball above the point of projection when thrown vertically upwards with an initial velocity of 15 ms^{-1} .

[2]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	$f(1) = 1 - 1 + 1 + 9 - 10 [= 0]$	B1	allow for correct division of $f(x)$ by $(x - 1)$ showing there is no remainder, or for $(x - 1)(x^3 + x + 10)$ found, showing it 'works' by multiplying it out	condone $1^4 - 1^3 + 1^2 + 9 - 10$
		i	attempt at division by $(x - 1)$ as far as $x^4 - x^3$ in working	M1	allow equiv for $(x + 2)$ as far as $x^4 + 2x^3$ in working or for inspection with at least two terms of cubic factor correct	eg for inspection, M1 for two terms right and two wrong

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	correctly obtaining $x^3 + x + 10$	A1	or $x^3 - 3x^2 + 7x - 5$ Examiner's Comments A large number of candidates successfully used the factor theorem to score the first mark and many went on to find the correct cubic factor - the majority of these choosing to do long division rather than use the inspection method. Some did not use the factor theorem but still showed that $x = 1$ was a root by successful division with no remainder. Those who used inspection without first applying the factor theorem did not in general show enough working for a convincing argument that there was no remainder and therefore that $x = 1$ was a root. A small number did not appear to understand what was meant by 'express $f(x)$ as the product of a linear factor and a quadratic factor' – some of these gained partial credit for the correct division seen in parts (ii) or (iii).	if M0 and this division / factorising is done in part (ii) or (iii), allow SC1 if correct cubic obtained there; attach the relevant part to (i) with a formal chain link if not already seen in the image zone for (i)
		ii	$[g(-2) =] -8 - 2 + 10$ or $f(-2) = 16 + 8 + 4 - 18 - 10$	M1	[in this scheme $g(x) = x^3 + x + 10$] allow M1 for correct trials with at least two values of x (other than 1) using $g(x)$ or $f(x)$ or $x^3 - 3x^2 + 7x - 5$ (may allow similar correct trials using division or inspection)	eg $f(2) = 16 - 8 + 4 + 18 - 10$ or 20 $f(3) = 81 - 27 + 9 + 27 - 10$ or 80 $f(0) = -10$ $f(-1) = 1 + 1 + 1 - 9 - 10$ or -16 No ft from wrong cubic 'factors' from (i)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$x = -2$ isw	A1	allow these marks if already earned in (i) Examiner's Comments Many used the correct method but made careless errors in calculations especially when trying negative values of x. Very few realised that they could use the factor theorem on the cubic they had found to obtain another root. Many confused 'root' with 'factor' and lost a mark.	NB factorising of $x^3 + x + 10$ or $x^3 - 3x^2 + 7x - 5$ in (ii) earns credit for (iii) [annotate with a yellow line in both parts to alert you – the image zone for (iii) includes part (ii)]
		iii	attempted division of $x^3 + x + 10$ by $(x + 2)$ as far as $x^3 + 2x^2$ in working	M1	or $x^3 - 3x^2 + 7x - 5$ by $(x - 1)$ as far as $x^3 - x^2$ in working or inspection with at least two terms of quadratic factor correct	alt method: allow M1 for attempted division of quartic by $x^2 + x - 2$ as far as $x^4 + x^3 - 2x^2$ in working, or inspection etc
		iii	correctly obtaining $x^2 - 2x + 5$	A1	allow these first 2 marks if this has been done in (ii), even if not used here	
		iii	use of $b^2 - 4ac$ with $x^2 - 2x + 5$	M1	may be in attempt at formula (ignore rest of formula)	or completing square form attempted or attempt at calculus or symmetry to find min pt NB M0 for use of $b^2 - 4ac$ with cubic factor etc
		iii	$b^2 - 4ac = 4 - 20 [= -16]$	A1	may be in formula;	or $(x - 1)^2 + 4$ or min = (1, 4)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	so only two real roots[of $f(x)$] [and hence no more linear factors]	A1	or no real roots of $x^2 - 2x + 5 = 0$; allow this last mark if clear use of $x^2 - 2x + 5 = 0$, even if error in $b^2 - 4ac$, provided result negative, but no ft from wrong factor if last M1 not earned, allow SC1 for stating that the only factors of 5 are 1 and 5 and reasoning eg that $(x - 1)(x - 5)$ and $(x + 1)(x + 5)$ do not give $x^2 - 2x + 5$ [hence $x^2 - 2x + 5$ does not factorise] Examiner's Comments Only about a third of the candidates found the correct quadratic factor. Those who found the quadratic usually gave sensible arguments based on the discriminant to show that only two real roots existed for the quartic. Some tried to use $b^2 - 4ac$ on the cubic $x^3 + x + 10$. Several candidates went back to square one and attempted to factorise the quartic rather than linking the earlier parts to the problem. Some candidates who had not progressed far in the first two parts sometimes made no attempt at this part.	or $(x - 1)^2 + 4$ is always positive so no real roots [of $(x - 1)^2 + 4 = 0$] [and hence no linear factors] or similar conclusion from min pt
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2			$4 + 2k + c = 0$ or $2^2 + 2k + c = 0$	B1	may be rearranged	
			$9 - 3k + c = 35$	B1	may be rearranged; the $(-3)^2$ must be evaluated / used as 9	condone -3^2 seen if used as 9
			correct method to eliminate one variable from their eqns	M1	eg subtraction or substitution for c ; condone one error	M0 for addition of eqns unless also multiplied appropriately
			$k = -6, c = 8$	A1	from fully correct method, allowing recovery from slips	if no errors and no method seen, allow correct answers to imply M1 provided B1B1 has been earned
			or			
			$[x^2 + kx + c =] (x - 2)(x - a)$	M1	or $(x - 2)(x + b)$	
			$-5 \times (-3 - a) = 35$ oe	M1		
			$a = 4$	A1		
			$k = -6, c = 8$	A1		
					Examiner's Comments Most candidates were able to make a start and substitute 2 and -3 into $f(x)$, although not all used the information given to write the results as equations. Errors in handling $(-3)^2$ or the 35 were common. Having obtained equations, many did not then go on to use standard methods to solve the simultaneous equations, or made errors in doing so. This meant that the full 4 marks available were given less often than examiners had hoped, although many picked up 2 or 3 marks.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	sketch of cubic the right way up, with two tps and clearly crossing the x axis in 3 places	B1		no section to be ruled; no curving back; condone slight 'flicking out' at ends but not approaching another turning point; condone some doubling (eg erased curves may continue to show); accept min tp on y-axis or in 3 rd or 4 th quadrant; curve must clearly extend beyond the x axis at both 'ends'
		i	crossing / reaching the x-axis at -4, -2 and 1.5	B1	intersections must be shown correctly labelled or worked out nearby; mark intent	accept curve crossing axis halfway between 1 and 2 if 3/2 not marked
		i	intersection of y-axis at -24	B1		NB to find -24 some are expanding f(x) here, which gains M1 in iiiA. If this is done, put a yellow line here and by (iii)A to alert you; this image appears again there
		i			Examiner's Comments Most candidates were able to sketch the correct shape for the cubic (the correct way up) and the majority were also able to correctly label the interceptions on the x-axis, although some gave the positive x intercept as $\frac{1}{2}$ or $\frac{2}{3}$ or 3. A few candidates failed to label the y-intercept or gave a wrong value such as 12 or -12. Some candidates drew their graph stopping at one of the roots (usually when $x = -4$) instead of crossing the x-axis. Only a small number of candidates drew the graph upside-down and a handful drew the wrong shape altogether.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	-2, 0 and $7/2$ oe isw or ft their intersections	2	B1 for 2 correct or ft or for $(-2, 0)$ $(0, 0)$ and $(3.5, 0)$ or M1 for $(x + 2) x (2x - 7)$ oe or SC1 for -6, -4 and $-1/2$ oe Examiner's Comments Quite a few errors were seen here, although a minority knew what to do and wrote down the correct values. Some gave factors or coordinates instead of roots, some solved $x - 2 = 0$ to give $x = 2$ as the root, and some went back to the equation but made an algebraic error in replacing x with $x - 2$, reaching $2x - 5$ as a factor instead of $2x - 7$.	
		iii	(A) correct expansion of product of 2 brackets of $f(x)$	M1	need not be simplified; condone lack of brackets for M1 or allow M1 for expansion of all 3 brackets, showing all terms, with at most one error: $2x^3 + 4x^2 + 8x^2 - 3x^2 + 16x - 12x - 6x - 24$	e.g. $2x^2 + 5x - 12$ or $2x^2 + x - 6$ or $x^2 + 6x + 8$ may be seen in (i) – allow the M1; the part (i) work appears at the foot of the image for (iii)A, so mark this rather than in (i)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	correct expansion of quadratic and linear and completion to given answer	A1	for correct completion if all 3 brackets already expanded, with some reference to show why -24 changes to -9 Examiner's Comments The first part was generally well done; most correctly expanded two brackets and continued to simplify and add 15 to get the required result. Common errors were: not dealing correctly with the 15 such as saying $g(x) = -15$ to get the result, and errors in expanding or collecting terms. There was some poor 'mathematical grammar' with the '+15' often appearing out of nowhere.	condone lack of brackets if they have gone on to expand correctly; condone '+15' appearing at some stage NB answer given; mark the whole process
		iii	(B) $g(1) = 2 + 9 - 2 - 9 [=0]$	B1	allow this mark for $(x - 1)$ shown to be a factor and a statement that this means that $x = 1$ is a root [of $g(x) = 0$] oe	B0 for just $g(1) = 2(1)^3 + 9(1)^2 - 2(1) - 9 [=0]$
		iii	attempt at division by $(x - 1)$ as far as $2x^3 - 2x^2$ in working	M1	or inspection with at least two terms of quadratic factor correct	M0 for division by $x + 1$ after $g(1) = 0$ unless further working such as $g(-1) = 0$ shown, but this can go on to gain last M1A1
		iii	correctly obtaining $2x^2 + 11x + 9$	A1	allow B2 for another linear factor found by the factor theorem	NB mixture of methods may be seen in this part – mark equivalently eg three uses of factor theorem, or two uses plus inspection to get last factor;
		iii	factorising a correct quadratic factor	M1	for factors giving two terms correct; eg allow M1 for factorising $2x^2 + 7x - 9$ after division by $x + 1$	allow M1 for $(x + 1)(x + 18/4)$ oe after -1 and $-18/4$ oe correctly found by formula

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$(2x + 9)(x + 1)(x - 1)$ isw	A1	allow $2(x + 9/2)(x + 1)(x - 1)$ oe; dependent on 2nd M1 only; condone omission of first factor found; ignore '= 0' seen Examiner's Comments In part (B) most candidates correctly showed $g(1) = 0$ although some failed to show enough working. Candidates were well-versed, in general, with the techniques of long division or inspection so that most achieved the correct quadratic factor and were able to go on and factorise this to gain full marks. Some tried to use the quadratic formula and then only gave $(x + 1)(x + 4.5)$ oe as factors.	SC alternative method for last 4 marks: allow first M1A1 for $(2x + 9)(x^2 - 1)$ and then second M1A1 for full factorisation
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	$\log_{10} p = \log_{10} a + \log_{10} 10^{kt}$	M1	condone omission of base;	if unsupported, B2 for correct equation
		i	$\log_{10} p = \log_{10} a + kt$ www	A1	Examiner's Comments The correct equation was often seen, but in many cases it stemmed from wrong working and didn't score. Some candidates stopped at $\log p = \log a + k \log 10$. $\log p = \log a \times kt$ was a common error; occasionally $\log p = \log a + k \log t$ or $\log p = \log a + \log kt$ were seen.	
		ii	2.02, 2.13, 2.23	B1	allow given to more sig figs	2.022304623 ..., 2.129657673, 2.229707433
		ii	plots correct	B1.f.t.	to nearest half square	
		ii	ruled line of best fit	B1	y-intercept between 1.65 and 1.7 and at least one point on or above the line and at least one point on or below the line Examiner's Comments This was done very well indeed, with just a few candidates making slips with the plots (usually the middle point), and a few joining each point with a ruler or drawing a curve of best fit to lose the last mark. Only a few candidates lost an easy mark by drawing their line of best fit freehand.	ft their plots must cover range from $x = 9$ to 49
		iii	0.0105 to 0.0125 for k	B1		must be connected to k
		iii	1.66 to 1.69 for $\log_{10} a$ or 45.7 to 49.0 for a	B1		must be connected to a

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\log_{10} p = \text{their } kt + \text{their } \log_{10} a$	B1	must be a correct form for equation of line and with their y-intercept and their gradient (may be found from graph or from table, must be correct method)	
		iii	$p = \text{their } "47.9 \times 10^{0.0115t}" \text{ or } 10^{1.6785 + 0.0115t}$	B1	as above, "47.9" and "0.0115" must follow from correct method Examiner's Comments Most were able to obtain values for the gradient and the y-intercept within the acceptable range, but not all knew what to do with these. For example, $\log 1.66$ or $10^{1.66}$ were often seen in the equation for $\log p$. A surprising number of candidates neglected to include an equation for $\log p$ at all, and went straight to an equation for p . This was sometimes correct, even if the equation for $\log p$ was incorrect. However, a common error was (for example) $p = 45.7 + 10^{0.012t}$.	
		iv	45.7 to 49.0 million	1	'million' needed, not just the value of p Examiner's Comments Although many candidates correctly identified the value of $\log a$ as crucial in their response, many of them neglected to include the word "million" and lost an easy mark.	
		v	reading from graph at 2.301..	M1*	or $\log_{10} 200 = " \log_{10} a + kt "$	or $200 = "10^{\log a + kt}"$ oe

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	their 54	M1dep*	eg for their $t = \frac{\log 200 - 1.68}{0.0115}$	or M1 for their $t = \frac{\log \frac{200}{47.9}}{0.0115}$
		v	2014 cao	A1	if unsupported, allow B3 only if consistent with graph Examiner's Comments Most candidates had the sense to revert to their graph. Accurate plotting and a good line of best fit often rewarded them with full marks. However, most candidates used their answer to part (iii) and sometimes lost the final mark due to rounding. A few used 200 000 000 instead of 200 in one of their equations and failed to score.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	$3x^2 - 6x - 22$	M1	condone one incorrect term, but must be three terms	condone "y ="
		i	their $y' = 0$ so	M1	at least one term correct in their y'	may be implied by use of e.g. quadratic formula, completing square, attempt to factorise
		i	3.89	A1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	-1.89	A1	if A0A0, SC1 for $\frac{3 \pm 5\sqrt{3}}{3}$ or $1 \pm \sqrt[5]{3}$ or better, or both decimal answers given to a different accuracy or from truncation Examiner's Comments Nearly all candidates differentiated successfully and set their derivative to zero. Over 60% of candidates went on to score full marks, although a few candidates made an error (usually $2x^2$ but occasionally $+ 24$ was retained). However, a significant minority attempted unsuccessfully to factorise the quadratic and then gave up and a surprising number were unable to use the quadratic formula correctly. Very few candidates appeared to check their answers. Some candidates lost an easy mark by leaving their answers in an exact form or by quoting a different precision. Occasionally, candidates found the second derivative and set this equal to zero. A significant minority wasted time either by finding the associated y-values or by determining the nature of the turning points, neither of which were required.	3.886751346 and -1.886751346
		ii	$x^3 - 3x^2 - 22x + 24 = 6x + 24$	M1	may be implied by $x^3 - 3x^2 - 28x [= 0]$	
		ii	$x^3 - 3x^2 - 28x [= 0]$	M1	may be implied by $x^2 - 3x - 28 [= 0]$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	other point when $x = 7$ isw	A1	dependent on award of both M marks <u>Examiner's Comments</u> This was very well answered by most candidates. Well over 80% earned the first mark and most went on to score full marks. Occasionally, candidates slipped up when collecting like terms and a few made a sign error when factorising. The minority who failed to score either omitted the question altogether, or set $6x + 24$ equal to the derivative.	ignore other values of x
		iii	$F[x] = \frac{x^4}{4} - \frac{3x^3}{3} - \frac{22x^2}{2} + 24x$	M1*	allow for three terms correct; condone $+ c$	alternative method M1 for $\int ((x^3 - 3x^2 - 22x + 24) - (6x + 24))dx$ may be implied by 2 nd M1
		iii	$F[0] - F[-4]$	M1dep	allow $0 - F[-4]$, condone $- F[-4]$, but do not allow $F[-4]$ only	M1* for $F[x] = \frac{x^4}{4} - \frac{3x^3}{3} - \frac{28x^2}{2}$ condone one error in integration
		iii	area of triangle = 48	B1		M1dep for $F[0] - F[-4]$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	area required = 96 from fully correct working	A1	A0 for – 96, ignore units, <u>Examiner's Comments</u> This question was accessible to most candidates, although a significant minority scored zero. Many candidates found the area of the triangle using $\frac{1}{2} \times \text{base} \times \text{height}$. Most of those who used a base of –4 realised that a negative area was impossible and so removed the minus sign. Some used integration and more often than not were successful – sometimes after 'losing' a minus sign. Most candidates also integrated successfully, but some made no further progress, as they ignored the upper limit and then 'airbrushed' the minus sign. A good proportion of those who did integrate successfully then made errors with the arithmetic. Some candidates earned two marks by combining the equations and integrating correctly, but a similar proportion ignored the upper limit or made arithmetical slips.	no marks for 96 unsupported
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	$\log_{10} h = \log_{10} a + bt$ www	B1	Examiner's Comments Wrong working often spoiled a correct final answer in this question. It was disappointing to see a significant proportion of candidates failing to score both marks on a very standard piece of work.	condone omission of base
		i	$m = b, c = \log_{10} a$	B1		must be clearly stated : linking equations is insufficient
		ii	-0.15, 0[.00], 0.23, 0.36, 0.56, 0.67, 0.78, 0.91, 1.08, 1.2[0]	B2	B1 if 1 error	no ft available for plots
		ii	plots correct (tolerance half square)	B1	condone 1 error – see overlay	
		ii	single ruled line of best fit for values of x from 5 to 50 inclusive	B1	line must not go outside overlay between $x = 5$ and $x = 50$ Examiner's Comments This was very well done. A few candidates made errors in the table — usually the first or the penultimate value. A tiny minority gave all values to a different degree of accuracy to the one requested, thus losing two easy marks — although credit was still available for the plots and the line. Most plotted the points adequately and drew a single ruled line of best fit across the whole range of x -values to earn two marks.	
		iii	$-0.3 \leq y\text{-intercept} \leq -0.22$	B1	may be implied by $0.5 \leq a \leq 0.603$	condone values from table; condone slips eg in reading from graph
		iii	valid method to find gradient of line	M1	may be embedded in equation; may be implied by eg m between 0.025 and 0.035	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$h = \text{their } a \times 10^{\text{their } bt}$ or $h = 10^{\text{their } \log a + \text{their } bt}$	M1	Examiner's Comments Those candidates who used their graph to find the gradient and the intercept often went on to score full marks in this part. Those who adopted other methods such as simultaneous equations often went astray, and obtaining a positive value for the Y-intercept or a large value for the gradient evidently did not cause concern. It would seem that a significant minority did not connect this part with earlier parts of the question..	if B1M1M0, then SC1 for $\log h = \log a + \text{their } bt$ is awarded if both values in the acceptable range for A1
		iii	$0.028 \leq b \leq 0.032$ and $0.5 \leq a \leq 0.603$ or $-0.3 \leq \log a \leq -0.22$	A1		
		iv	$a10^{60b} - a10^{50b}$ their values for a and b	M1	or $10^{\log a + b \times 60} - 10^{\log a + b \times 50}$ or their values for $\log a$ and b	condone 15.9 as second term may follow starting with $\log h = \log a + \text{their } bt$
		iv	8.0 to 26.1 inclusive	A1	Examiner's Comments $t = 70, 10$ and 55 were all seen, but many candidates used $t = 60$ successfully with their model, and then subtracted either 15.9 or $f(50)$ to earn both marks. Unfortunately a few candidates stopped at $f(60)$ lost both marks.	NB A0 for estimate without clear valid method using model; both marks available even if a or b or both are outside range in (iii)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	comment on the continuing reduction in thickness and its consequences	B1	eg in long term, it predicts that reduction in thickness will continue to increase, even when the glacier has completely melted Examiner's Comments Many candidates wrote sensible and worthy responses to this question. Unfortunately, many of them failed to score, in spite of their likely truth, as they were vague or missed the point. Candidates were expected to comment on the model continuing to predict an ever increasing rate of reduction in the thickness of the ice, in spite of the fact that at some point all the ice will have melted.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	$\log_{10} y = \log_{10} a + bt$ www	B1	B0 for just $\log_{10} y = \log_{10} a + bt \log_{10} 10$	allow omission of base throughout question
		i	gradient is b , intercept is $\log_{10} a$ cao	B2	B1 for one correct; award independently of their equation; must be stated – linking by arrows etc is insufficient; condone $m = b$ and $c = \log a$ Examiner's Comments Many scored full marks in this part, but of those who derived the equation, a significant minority did so incorrectly, thus losing the first mark. “ bt ” was sometimes quoted as the gradient, and “ $a = \text{intercept}$ ” was a common error. Some candidates failed to state the gradient or the intercept, simply drawing lines to their equation or linking with $y = mx + c$. This is insufficient.	ignore t -intercept is $\frac{-\log_{10} a}{b}$ B0 for gradient is bt
		ii	1.58, 1.8[0], 1.98, 2.37, 2.68	B1	allow values which round to these numbers to 2 dp;	all values must be correct
		ii	all values correct and all plotted accurately	B1	within tolerance on overlay;	
		ii	ruled line of best fit for at least $1 \leq t \leq 10$	B1	within tolerance on overlay: must not cut red or green line; line between (1, 0.6) and (1, 1.05) at lower limit and between (10, 2.3) and (10, 2.75) at upper limit;	use ruler tool to check if line is ruled where necessary; tolerance: one small square horizontally at each end; not dependent on correct plots

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	evaluation of $\frac{\log y_2 - \log y_1}{t_2 - t_1}$ or substitution of $(t_1, \log y_1)$ and $(t_2, \log y_2)$ in $\log y = bt + \log a$ to obtain a numerical value for the gradient	M1	$(t_1, \log y_1)$ and $(t_2, \log y_2)$ are points on their line	condone use of values from table
		ii	$0.14 \leq b \leq 0.24$	A1	gradient must be identified as b for A1	
		ii	$2.5 \leq a \leq 6.3$	B1	must be identified as a ; not from wrong working	
		ii	$y = \text{their } a \times 10^{\text{their } b \times t} \text{ or } y = 10^{\text{their } bt + \text{their } \log a} \text{ or } 10^{\text{their } \log a} \times 10^{\text{their } b \times t} \text{ oe}$	M1		if M0A0B0M0 allow SC3 for substitution directly into given formula to obtain $y = a10^{bt}$ with a and b in acceptable range
		ii	a and b or $\log a$ and b both in acceptable range	A1	$0.4 \leq \log a \leq 0.8$ Examiner's Comments Most completed the table successfully, and went on to plot the points and draw a suitable line of best fit. A few lost an easy first mark through poor calculator skills (2.34 instead of 2.37 was quite common) and some rounded to 1 decimal place. A few candidates drew a curve of best fit, or failed to use a ruler. Most were able to find the gradient of the line for an easy mark, but many failed to link this to b . Similarly, the instruction to find the value of a was often disregarded. Surprisingly, many candidates simply stopped when they had found a and b , thus losing the last two marks.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	260 or 261	B1	B0 for non-integer answer Examiner's Comments The majority of candidates successfully obtained the correct value, but a significant minority lost an easy mark by failing to give the answer in context as an integer.	
			Total	12		
8			$1 < x < 3 \Rightarrow -1 < x - 2 < 1$ $\Rightarrow x - 2 > 1$	B1 B1	oe [or $a = 2$ and $b = 1$] Examiner's Comments The non-standard nature of the question made this one of the harder section A questions. Some candidates were able to write the answer down while others used an algebraic approach.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	Range is $-1 \leq y \leq 3$	M1	-1, 3	
		i		A1	$-1 \leq y \leq 3$ or $-1 \leq f(x) \leq 3$ or $[-1, 3]$ (not -1 to 3 , $-1 \leq x \leq 3$, $-1 < y < 3$ etc) <u>Examiner's Comments</u> This was generally well done, with one mark awarded for -1 and 3 seen, and one for the correct notation. Some used x instead of y or $f(x)$, and others confused domain and range.	
		ii	$y = 1 - 2\sin x \Leftrightarrow y$		[can interchange x and y at any stage]	
		ii	$x = 1 - 2\sin y \Rightarrow x - 1 = -2 \sin y$	M1	attempt to re-arrange	
		ii	$\Rightarrow \sin y = (1 - x)/2$	A1	o.e. e.g. $\sin y = (x - 1)/(-2)$ (or $\sin x = (y - 1)/(-2)$)	
		ii	$\Rightarrow y = \arcsin [(1 - x)/2]$	A1	or $f^{-1}(x) = \arcsin [(1 - x)/2]$, not x or $f^{-1}(y) = \arcsin[1 - y]/2$ (viz must have swapped x and y for final 'A' mark) $\arcsin [(x - 1)/-2]$ is A0 <u>Examiner's Comments</u> Most candidates are well practiced at finding inverses, and were familiar with arcsine, gaining full marks here. Leaving the result as $y = \arcsin((x - 1)/-2)$ lost the final A1. Very occasionally, candidates gave the answer as $1/f(x)$ or $f'(x)$.	
		iii	$f'(x) = -2\cos x$	M1	condone $2\cos x$ ☐	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow f'(0) = -2$	A1	cao	
		iii	$\Rightarrow$ gradient of $y = f^{-1}(x)$ at $(1, 0) = -\frac{1}{2}$	A1	not $1/-2$ <u>Examiner's Comments</u> Nearly all candidates found $f'(x)$ and $f'(0)$ correctly. The gradient of the inverse function was less successful. Many confused this with the condition for perpendicularity and gave the answer $\frac{1}{2}$ instead of $-\frac{1}{2}$. Those who tried to differentiate $f^{-1}(x)$ directly had little success.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	$a = \frac{1}{2}$	B1	or 0.5	
		i	$b = 1$	B1	<u>Examiner's Comments</u> Some candidates were able to write down the correct values of a and b . Those who chose to use transformation arguments sometimes confused the stretch ($\frac{1}{2}$ or 2) and the translation ($+1$ or -1). Others chose to substitute the coordinates of specific points, with variable success.	
		ii	$\frac{1}{2} x + 1 = x $ $\Rightarrow \frac{1}{2} (x + 1) = x,$	M1	o.e. ft their a ($\neq 0$), b (but allow recovery to correct values) or verified by subst $x = 1, y = 1$ into $y = \frac{1}{2} x + 1 $ and $y = x $	
		ii	$\Rightarrow x = 1, y = 1$	A1	unsupported answers M0A0	
		ii	or $\frac{1}{2}(x + 1) = -x,$	M1	o.e., ft their a, b ; or verified by subst $(-1/3, 1/3)$ into $y = \frac{1}{2} x + 1 $ and $y = x $	
		ii	$\Rightarrow x = -1/3, y = 1/3$	A1	or 0.33, -0.33 or better unsupported answers M0A0	
		ii	or $\frac{1}{4}(x + 1)^2 = x^2$	M1	ft their a and b	
		ii	$\Rightarrow 3x^2 - 2x - 1 = 0$	M1ft	obtaining a quadratic = 0, ft their previous line, but must have an x^2 term	
		ii	$\Rightarrow x = -1/3$ or 1	A1	SC3 for $(1, 1)$ $(-1/3, 1/3)$ and one or more additional points	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y = 1/3$ or 1	A1	<u>Examiner's Comments</u> Most candidates, who knew what they were doing here either used $\frac{1}{2}(x+1) = \pm x$ or squared both sides to find a quadratic in x. In the latter approach, some forgot to square the $\frac{1}{2}$ and got the wrong quadratic. Examiners followed through their values for a and b. Some candidates omitted the y-coordinates. Candidates who found $(1, 1)$ without showing a valid method got no marks, and there was evidence of the usual mistakes in using modulus, such as $x+1 = x + 1$, etc.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11		i	$a = 2, b = \frac{1}{2}$	B1B1	Examiner's Comments For some, this was a write-down for 2 marks. Some used transformation arguments, other substituted in particular coordinates. The most common errors were $a = 3$ and $b = 2$ or $\frac{1}{4}$.	
		ii	$y = 2 + \cos \frac{1}{2} x \Leftrightarrow x \Leftrightarrow y$ $x = 2 + \cos \frac{1}{2} y$		(may be seen later)	
		ii	$\Rightarrow x - 2 = \cos \frac{1}{2} y$	M1	subtracting [their] a from both sides (first)	need not substitute for a, b
		ii	$\Rightarrow \arccos(x - 2) = \frac{1}{2} y$	M1	$\arccos(x - [\text{their}] a) = [\text{their}] b \times y$	or with $x \Leftrightarrow y$, need not subst for a, b
		ii	$\Leftrightarrow y = f^{-1}(x) = 2 \arccos(x - 2)$	A1	cao or $2 \cos^{-1}(x - 2)$	may be implied by flow diagram
		ii	Domain $1 \leq x \leq 3$	M1	domain 1 to 3, range 0 to 2π	if not stated, assume first is domain
		ii	Range $0 \leq y \leq 2\pi$	A1	correctly specified: must be $\leq, x$ for domain, y or f^{-1} or $f^{-1}(x)$ for range	allow $[1, 3], [0, 2\pi]$ not 360° (not f)
					Examiner's Comments We allowed plenty of follow-through marks for the first two method marks here, which were almost always gained. Fully correct inverse functions were common. As for domain and range, most seem to know that these are the reverse of the domain and range for f. However, the 'A' mark here demanded accurate use of notation, with 'x' used for domain and 'y' for range, and this mark was often lost.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	7		
12			$ 3 - 2x = 4 x $ $\Rightarrow 3 - 2x = 4x, x = \frac{1}{2}$ or $3 - 2x = -4x, x = -1\frac{1}{2}$ <i>or</i> $(3 - 2x)^2 = 16x^2$ $\Rightarrow 12x^2 + 12x - 9 = 0$ $\Rightarrow x = \frac{1}{2}, -1\frac{1}{2}$	M1A1 M1A1 M1 A1 A1 A1	not $3/(3 - 2)$ squaring both sides correct quadratic o.e. but with single x^2 term Examiner's Comments Although plenty of candidates scored full marks with apparent ease, there were all sorts of errors as well. Some clearly do not understand the modulus function; many duplicate work by solving 4 equations from $\pm (3 - 2x) = \pm 4x$, and in the process produced additional solutions due to poor algebra. A surprisingly common error was to write $3 = 6x \Rightarrow x = 2$! Some even discounted the solution $x = -3/2$ on the grounds that answers to a modulus question need to be positive! Squaring both sides was seen occasionally, and although this method is somewhat long-winded, it does avoid conceptual errors such as $ 3 - 2x = 3 + 2x$.	If 3 or more final answers offered, – 1 for each incorrect additional answer – 1 for final ans written as an inequality $(3 - 2x)^2 = 4x^2$ is M0
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$ff(x) = f\left(\frac{1-x}{1+x}\right) = \frac{1 - \frac{1-x}{1+x}}{1 + \frac{1-x}{1+x}}$	M1	substituting $(1-x)/(1+x)$ for x in $f(x)$	
		i	$= \frac{1+x-1+x}{1+x+1-x} = \frac{2x}{2} = x^*$	A1	correctly simplified to x NB AG	
		i	$^{-1}(x) = f(x) = (1-x)/(1+x)$	B1	or just $f^{-1}(x) = f(x)$	
Examiner's Comments Most candidates gained a method mark for substituting $(1-x)/(1+x)$ for x in $f(x)$. However, the simplification of the ensuing algebraic fraction proved to be problematic to many candidates, who failed to clear the subsidiary denominators correctly. Concluding that $f^{-1}(x) = f(x)$ should of course be a 'write down' from $ff(x) = x$; however, virtually all candidates found $f^{-1}(x)$ by rearranging the formula for $x = f(y)$, usually correctly. Occasionally we were offered $f^{-1}(x) = 1/f(x) = (1+x)/(1-x)$.						
		ii	$g(-x) = \frac{1-(-x)^2}{1+(-x)^2}$	M1	substituting $-x$ for x in $g(x)$ condone use of 'f' for g	if brackets are omitted or misplaced allow M1A0
		ii	$= \frac{1-x^2}{1+x^2} = g(x)$	A1	must indicate that $g(-x) = g(x)$ somewhere	condone use of 'f' for g

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Graph is symmetrical about the y -axis.	B1	allow 'reflected', 'reflection' for symmetrical Examiner's Comments This was well answered, with few candidates using particular values of x to 'show' that $g(x)$ was even. We condoned the use of f instead of g. Occasionally the brackets were misplaced in $1 - (-x)^2$ or $1 + (-x)^2$. The geometrical interpretation was well answered: although we would prefer 'symmetrical about the y-axis' to formulations such as 'reflection in the y-axis', the latter was nevertheless condoned.	must state axis (y -axis or $x = 0$)
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$1 + k$	B1		Examiner's Comments The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. This was an easy write-down for virtually all candidates, except those few who did not know that $e^0 = 1$.
		ii	$f'(x) = 2e^{2x} - 2ke^{-2x}$	B1		
		ii	$f'(x) = 0 \Rightarrow 2e^{2x} - 2ke^{-2x} = 0$	M1	their derivative = 0	
		ii	$\Rightarrow e^{2x} = ke^{-2x}$			
		ii	$\Rightarrow e^{4x} = k, 4x = \ln k, x = \frac{1}{4} \ln k$	A1	NB AG	
		ii	$y = e^{(\frac{1}{2} \ln k)} + ke^{(-\frac{1}{2} \ln k)}$	M1	substituting $x = \frac{1}{4} \ln k$ into $f(x)$	

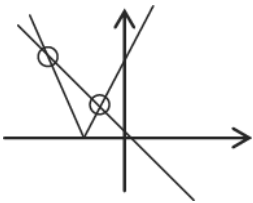
Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \sqrt{k} + k/\sqrt{k} = 2\sqrt{k}$	A1cao	or $2k^{1/2}$	Examiner's Comments The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The first two marks were pretty universally earned, but deriving $x = \frac{1}{4} \ln k$, together with the final 'A' mark for getting $2\sqrt{k}$, caused a few problems, with some inaccurate logarithm work. For example, $e^{1/2 \ln k} = 1/2 k$ was a commonly seen misconception.
		iii	$\text{Area} = \int_0^{\frac{1}{2} \ln k} (e^{2x} + ke^{-2x}) [dx]$	B1	correct integral and limits (soi)	
		iii	$= \int_0^{\frac{1}{2} \ln k} (e^{2x} + ke^{-2x}) dx = \left[\frac{1}{2} e^{2x} - \frac{1}{2} ke^{-2x} \right]_0^{\frac{1}{2} \ln k}$	B1	$\left[\frac{1}{2} e^{2x} - \frac{1}{2} ke^{-2x} \right]$	
		iii	$= \frac{1}{2} k - \frac{1}{2} - \frac{1}{2} + \frac{1}{2} k$	M1	$e^{\ln k} = k$ or $e^{-\ln k} = 1/k$ (soi)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= k - 1$	A1		Examiner's Comments The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The integration was usually correct, but, thereafter, as in part (ii), the simplification to arrive at $k - 1$ proved to be tricky, with similar errors being made.
		iv	(A) $g(x) = e^{2(x + \frac{1}{4} \ln k)} + k e^{-2(x + \frac{1}{4} \ln k)}$	M1	Substitute $x + \frac{1}{4} \ln k$ for x in $f(x)$	condone missing brackets
		iv	$= e^{2x} \cdot e^{\frac{1}{2} \ln k} + k e^{-2x} \cdot e^{-\frac{1}{2} \ln k}$	M1	$e^{p+q} = e^p \times e^q$ used	
		iv	$= (e^{\ln k})^{1/2} e^{2x} + k \cdot (e^{\ln k})^{-1/2} e^{-2x}$			
		iv	$= k^{1/2} e^{2x} + k \cdot k^{-1/2} \cdot e^{-2x}$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$= \sqrt{k} (e^{2x} + e^{-2x})^*$	A1	NB AG – must show enough working	e.g. $k^{1/2} e^{2x} + k \cdot k^{-1/2} \cdot e^{-2x}$ Examiner's Comments The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx} , but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. Most attempts correctly substituted $x + \frac{1}{4} \ln k$ for x in $f(x)$ to gain the first M mark, but we needed to see clear evidence of how this simplifies to the given result. Often candidates seemed to be working backwards from this without really understanding the process.
		iv	$(B) g(-x) = \sqrt{k} (e^{-2x} + e^{2x})$	M1	substituting $-x$ for x	condone 'f' used instead of 'g' for M1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$= g(x)$ so g is even	A1	must include $g(-x) = g(x)$, and either define an even function or conclude that g is even	not $f(-x) = f(x)$ for A1 Examiner's Comments The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The definition of an even function was well known, but sometimes the structuring of the proof was indecisively presented. Some used 'f' instead of 'g' (here, f is indeed not an even function!), and we required to see either a clear statement of the definition of an even function, or a clear conclusion that g is therefore even. The structure '$g(-x) = \dots = \dots = g(\dots) \Rightarrow g$ is even' is the most transparent formulation to use in such proofs, rather than starting them by stating that $g(-x) = g(x)$, viz the result they are trying to prove!
		iv	(C) $g(x)$ is symmetrical about the y -axis, and	B1		
		iv	$f(x)$ is $g(x)$ translated $\frac{1}{4} \ln k$ in x -direction	B1	allow 'shift' or 'move'	or g is f translated $-\frac{1}{4} \ln k$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	so $f(x)$ is symmetrical about $x = \frac{1}{4} \ln k$	B1	allow final B1 even if unsupported	or incorrectly supported Examiner's Comments The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx} ; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The argument here proved beyond most candidates, with only 20% getting full marks. Many stated that f was an even function, perhaps thinking that any line of symmetry sufficed. Sometimes it was indeed a little difficult to decide whether candidates were referring to f or g in their answers.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15			 $x = -1$ $x = -1/3$	M1 A1 B1 B1	Sketch of $y = 2x + 1 $ $y = -x$ and two intersections indicated not from ww, condone $(-1, 1)$ not from ww, condone $(-1/3, 1/3)$	condone no intercept labels, but must be a 'V' shape with vertex on $-ve$ x axis squaring: $(2x + 1)^2 = x^2 \Rightarrow 3x^2 + 4x + 1 = 0$ $\Rightarrow (3x + 1)(x + 1) = 0, x = -1, -1/3$ Examiner's Comments Sketches of the modulus function with $y = -x$ were generally well done, though quite a few lost a mark for neither clearly indicating the intercepts nor making a clear statement that there were two of them. The roots were then usually found correctly, with less evidence of faulty modulus algebra than in recent years.
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16			$fg(x) = \ln(2 + e^x)$ $\Rightarrow \ln(2 + e^x) = 2x$ $\Rightarrow 2 + e^x = e^{2x}$ $\Rightarrow e^{2x} - e^x - 2 [= 0]$ $\Rightarrow (e^x - 2)(e^x + 1) = 0, e^x = 2, -1$ $\Rightarrow e^x = 2, x = \ln 2$	M1 A1 M1 A1 A1	condone missing brackets Rearranging into a quadratic in e^x obtaining roots 2, -1 $x = \ln 2$ only, not from ww	may be implied from both correct roots -1 root may be inferred from factorising $x = \ln(-1)$ is A0 Examiner's Comments Virtually all candidates formed the composite function in the correct order to obtain $fg(x) = \ln(2 + e^x)$. A few then simplified this to $\ln 2 + x$ and therefore made no further progress. Of those who did correctly proceed to $2 + e^x = e^{2x}$, a substantial minority then incorrectly took logs of each side to reach $\ln 2 + x = 2x$. Of those who correctly rearranged the equation into a quadratic in e^x , nearly all then gained full marks, correctly rejecting the $e^x = -1$ solution.
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	(A) Distance travelled = Area under the graph	M1	Attempt to find area	
		i	$\frac{1}{2} \times 4 \times 8 + \frac{1}{2} \times 4 \times (8 + 12) + 4 \times 12$	M1	Splitting into suitable parts	
		i	104 m	A1	Cao	
					Allow all 3 marks for 104 without any working	
		i	(B) Either Working backwards from distance when $t = 12$	M1		
		i	$12 - \frac{(104 - 100)}{12}$	M1	Allow this mark for 0.33... Follow through from their total distance	
		i	11.67 s	A1	Cao	
		i	Or			
		i	Working forwards from when $t = 8$	M1		
		i	$8 + \frac{(100 - 56)}{12}$	M1	Allow this mark for 3.67... Follow through from their distance at time 8s	
		i	11.67 s	A1	Coa	
		ii	Substituting $t = 8$ gives $v = \frac{5}{2} \times 8 - \frac{1}{8} \times 8^2 = 12$	B1		
		iii	Distance $= \int_0^{12} \left(\frac{5t}{2} - \frac{t^2}{8} \right) dt$	M1	Integrating v. Condone no limits.	
		iii	$\left[\frac{5t^2}{4} - \frac{t^3}{24} \right]_0^{12}$	A1	Condone no limits	
		iii	[180-72] (-[0])	M1	Substituting $t = 12$	
		iii	108 m	A1		
		iv	Model P: distance at $t = 11.35$ is 96.2	B1	Cao	
		iv	Model Q: distance at $t = 11.35$ is			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$\left[\frac{5t^2}{4} - \frac{t^3}{24} \right]_0^{11.35} = 100.1$	M1	Substituting 11.35 in their expression from part (iii)	
		iv	Model Q places the runner closer	E1	Cao from correct previous working for both models	
		v	Model P: Greatest acceleration $\frac{8}{4} = 2 \text{ m s}^{-2}$	B1		
		v	Model Q: $a = \frac{dv}{dt} = \frac{5}{2} - \frac{t}{4}$	M1	Differentiating v	
		v		A1		
		v	Model Q: Greatest acceleration is 2.5 ms^{-2}	B1	Award if correct answer seen	
					Examiner's Comments This was the first of the two Section B questions. It involved two models for the speed of a runner covering 100 metres. It was well answered. In part (i) candidates worked from a given speed-time graph and most were successful in doing so; however some did not realise that the second request needed some calculation and could not be obtained just from reading off the graph. The question then presented the second model as an equation for v in terms of t; most candidates realised that the questions on this involved the use of calculus and answered them correctly. The last two parts of the question involved comparing results from the two models and there were many correct answers, well presented with clear statements as to which model was being considered.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			Total	18			
18			$y = x^3 - 4$ $x \ll y$ $x = y^3 - 4$ $\Rightarrow x + 4 = y^3$ $\Rightarrow y = \sqrt[3]{x+4}$ so $f^{-1}(x) = \sqrt[3]{x+4}$ range of f is $-5 \leq y \leq 4$ so domain of f^{-1} is $-5 \leq x \leq 4$ range is $-1 \leq y \leq 2$	M1(AO1.1) A1(AO1.1) M1(AO1.1) A1(AO1.2) B1(AO1.1) [5]	attempt to invert accept $y = \sqrt[3]{x+4}$ but not $x = \sqrt[3]{y+4}$ May be implied or $[-5, 4]$ or $-1 \leq f^{-1}(x) \leq 2$ or $[-1, 2]$		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	$\int \frac{dm}{m} = \int \frac{dt}{t(1+2t)}$ $\frac{1}{t(1+2t)} \equiv \frac{A}{t} + \frac{B}{1+2t}$ $\Rightarrow 1 \equiv A(1+2t) + Bt$ $t=0 \Rightarrow A=1$ $t=-\frac{1}{2} \Rightarrow 1 = -\frac{1}{2}B \Rightarrow B=-2$ $\Rightarrow \int \frac{dm}{m} = \int \left(\frac{1}{t} - \frac{2}{1+2t} \right) dt$ $\Rightarrow \ln m = \ln t - \ln(1+2t) + c$ $t=1, m=1 \Rightarrow c = \ln 3$ $\Rightarrow \ln m = \ln \left(\frac{3t}{1+2t} \right)$ $\Rightarrow m = \frac{3t}{(1+2t)}$	M1(AO1.1a) M1(AO3.1b) M1(AO1.1) A1A1(AO1.1.1.1) B1FT(AO2.1) M1(AO1.1) E1(AO2.1) [8]	separating variables using partial fractions substituting values, equating coeffs or cover up $A=1, B=-2$ FT their i, ii , condone no c evaluating constant of integration AG		
		b	i $1.25 = \frac{3t}{(1+2t)}$ $\Rightarrow 1.25 + 2.5t = 3t$ $\Rightarrow t = 1.25 \div 0.5 = 2.5$ minutes	M1(AO1.1a) A1(AO1.1) [2]	Enter text here.		
		b	ii $m = \frac{3}{\left(\frac{1}{t} + 2\right)}$ $\rightarrow 1.5$ [grams]	M1(AO3.1b) A1(AO2.2a) [2]	Enter text here.		
			Total	12			

Question		Answer/Indicative content	Marks	Part marks and guidance		
20		Either $ 2x - 1 \geq 4$ $\Rightarrow \frac{2x - 1}{4} \geq$ or $2x - 1 \leq -4$ Or $(2x - 1)^2 \geq 16$ $4x^2 - 4x - 15 \geq 0$ $(2x - 5)(2x + 3) \geq 0$ $\Rightarrow x \geq 2\frac{1}{2} \quad \Rightarrow x \leq -1\frac{1}{2}$ $\{x : x \leq -1\frac{1}{2}\} \cup \{x : x \geq 2\frac{1}{2}\}$	M1(AO1.1) M1(AO1.1) M1(AO1.1) M1(AO1.1) M1(AO1.1) A1(AO1.1) A1(AO2.5) [4]	M1 for sketch graph of $y = (2x - 1)^2$ and $y = 16$ M1 for $x = 2\frac{1}{2}, -1\frac{1}{2}$ OR $x \geq 2\frac{1}{2}$ or $x \leq -1\frac{1}{2}$ If final ans not in one of these forms then withhold final A1		
		Total	4			



Question			Answer/Indicative content	Marks	Part marks and guidance	
22		a	$[384 - 400 + 8 + 8 =] 0$	B1(AO1.1) [1]		
		b	Long division or equating coefficients $6x^2 - x - 2$ seen $(x - 4)(3x \pm 2)(2x \pm 1)$ $(x - 4)(3x - 2)(2x + 1)$	M1(AO2.1) A1(AO1.1) M1(AO1.1) A1(AO1.1) [4]	DR	
			Total	5		

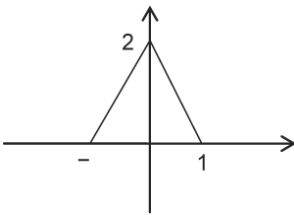
Question			Answer/Indicative content	Marks	Part marks and guidance		
23		a	$(1-2x)^{-\frac{1}{2}} \approx 1 + \left(-\frac{1}{2}\right)(-2x) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-2x)^2 + \dots$ $= 1 + x + \frac{3}{2}x^2 \quad \text{AG}$	M1(AO 1.1b) A1(AO 2.1) [2]	Correct form required; allow sign errors		
		b	Valid when $ x < \frac{1}{2}$	E1(AO 2.3) [1]			
		c	$(1+2x)^{\frac{1}{2}} = 1 + x - \frac{1}{2}x^2 + \dots$ $\left(1 + x - \frac{1}{2}x^2\right)\left(1 + x + \frac{3}{2}x^2\right)$ $= 1 + 2x + 2x^2$ Alternative method $\sqrt{\frac{1+2x}{1-2x}} = \sqrt{\frac{(1+2x)(1+2x)}{(1-2x)(1+2x)}} = \frac{1+2x}{\sqrt{1-4x^2}}$ $= (1+2x)\left(1 + \left(-\frac{1}{2}\right)(-4x^2) + \dots\right)$ $= 1 + 2x + 2x^2$	B1(AO 1.1a) M1(AO 1.1a) A1(AO 1.1b) Alternative method M1 M1 A1 [3]	Product of their expansions attempted		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	$\sqrt{\frac{1.1}{0.9}} = \frac{\sqrt{11}}{3} \approx 1 + 2 \times \frac{1}{20} + 2 \times \left(\frac{1}{20}\right)^2$ $\sqrt{11} \approx 3.315$	M1(AO 2.1) A1(AO 2.2a) [2]	Obtaining an expression involving $\sqrt{11}$ Or $\frac{663}{200}$		
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance		
24		a	$[g(x) =] x $	B1(AO 1.1) [1]			
		b	All non-negative real numbers	B1(AO 1.1) [1]	Allow any reasonable notation, e.g. $y \geq 0$		
		c	Either $ 3x - 1 > 1$ $\Rightarrow 3x - 1 > 1$ or $3x - 1 < -1$ Or $(3x - 1)^2 > 1$ $9x^2 - 6x > 0 \Rightarrow 3x(3x - 2) > 0$ Hence $x > \frac{2}{3}, x < 0$ $\{x : x < 0\} \cup \{x : x > \frac{2}{3}\}$	M1(AO 1.1) M1(AO 1.1) M1 M1 A1(AO 1.1) A1(AO 2.5) [4]	Allow for sketch of $y = (3x - 1)^2$ and $y = 1$ Allow for critical values $x = 0, \frac{2}{3}$ For either of these inequalities Final answer must be in this set notation form or else stated '$x < 0$ or $x > \frac{2}{3}$', as		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
25			DR Substitute $x = 0$ or $y = 0$ in $2x + y = 6$ Line crosses axes at $(0, 6)$ and $(3, 0)$ Quadratic with factor $(x - 3)$ Repeated factor $(x - 3)$ $y = \frac{2}{3}(x - 3)^2$ oe	M1(AO 3.1a) A1(AO 1.1) M1(AO 3.1a) M1(AO 1.1) A1(AO 2.1) [5]	Must follow from clear reasoning		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
26			DR $\frac{b^2 - 13}{1 - \frac{1}{b}} = -6$ $\frac{b(b^2 - 13)}{b - 1} = -6$ $b(b^2 - 13) = -6(b - 1)$ $b^3 - 7b - 6 = 0$ $(-1)^3 - 7 \times (-1) - 6 = 0$ $(b + 1)(b^2 - b - 6) = 0$ $(b + 1)(b + 2)(b - 3) = 0$ Roots $-1, -2, 3$ -1 cannot be the common ratio of a geometric sequence with a sum to infinity Possible common ratios are $-\frac{1}{2}$ and $\frac{1}{3}$	M1(AO 3.1a) M1(AO 3.1a) A1(AO 1.1) M1(AO 1.1) M1(AO 2.1) A1(AO 1.1) M1(AO 1.1) B1(AO 2.3) A1FT(AO 3.2a) [9]	Use of sum to infinity to form an equation Starting to clear fractions Correct equation without fractions Cubic with zero on one side Use of factor theorem to search for factor Correct fact Method for solving quadratic Rejection of root that does not make sense in the context Follow through their values of b		
			Total	9			

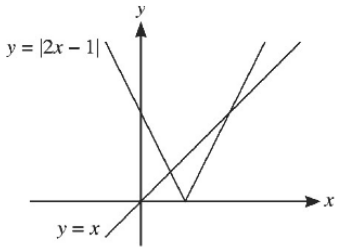
Question			Answer/Indicative content	Marks	Part marks and guidance		
27				M1 A1 A1 [3]	inverted 'v' shape through $(-1, 0)$, $(1, 0)$ and $(0, 2)$ correct domain $(-1 \leq x \leq 1)$ Examiner's Comments Only a third of candidates scored all three marks here. The final mark required the domain of the graph to be correct – often the 'v' shape extended beyond $x = -1$ to 1. Other attempts bore no relation to the correct answer.		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
28		i	$y = \ln(1 - x) \quad x \leftrightarrow y$ $x = \ln(1 - y)$ $\Rightarrow e^x = 1 - y$ $\Rightarrow y = 1 - e^x$ [so $f^{-1}(x) = 1 - e^x$] domain $x < \ln 2$ range $-1 < y < 1$	M1 A1 B1 B1 [4]	or $e^y = 1 - x$ $y = 1 - e^x$ or $f^{-1}(x) = 1 - e^x$ allow $x < 0.693$ or better, $-\infty < x < \ln 2$ or $-1 < f^{-1}(x) < 1$. Must use x for domain, y or $f^{-1}(x)$ for range.	can interchange x and y at any stage $x \leq \ln 2$ is B0, $\ln 0 < x < \ln 2$ is B0 allow $(-1, 1)$ but not $[-1, 1]$. If not labelled, take inequality with x as domain and with y or $f^{-1}(x)$ as range	Examiner's Comments The first two marks for finding the inverse function were nearly always gained. However, accurate notation for the domain and the range was not often seen. Not many candidates scored full marks, with the domain proving particularly awkward to get right.

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$f(-x) = \ln(1+x)$ $fg(x) = \ln(1-x^2)$ $\ln(1-x) + \ln(1+x) = \ln(1-x)(1+x)$ $= \ln(1-x^2)$	B1 B1 B1 [3]	soi e.g. from $\ln(1-x) + \ln(1+x) = \dots$	must include brackets must include brackets	
			Total	7			
29		a	$f(3) = 6 \times 27 - 17 \times 9 - 5 \times 3 + 6 = 0$	B1(AO1.1b) [1]	use of Factor theorem with convincing detail		
		b	$(x-3)(6x^2+x-2)$ valid attempt to factorise trinomial $(x-3)(2x-1)(3x+2)$	M1(AO1.1) A1(AO1.1b) M1(AO1.1b) A1(AO1.1b) [4]	allow sign errors in trinomial	or attempt at long division	
			Total	1			

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		a	At stationary point: $4x^3 - 9x^2 + 6x = 0$ $x(4x^2 - 9x + 6) = 0 \Rightarrow x = 0$ or $4x^2 - 9x + 6 = 0$ Discriminant of quadratic is $(-9)^2 - 4 \times 24 = -15 < 0$ So $x = 0$ gives the only stationary point $f(0) = 0^4 - 3 \times 0^3 + 3 \times 0^2 = 0$, so the point is the origin $f''(x) = 12x^2 - 18x + 6 \Rightarrow f''(0) = 6 > 0$ So the origin is a minimum point	M1(AO1.1a) M1(AO1.1a) M1(AO1.1a) A1(AO2.2a) B1(AO2.1) M1(AO1.1a) A1(AO2.2a) [7]	Attempt to differentiate & equate to 0 Factorising the cubic oe (quadratic formula, completing the square, complex roots from calculator) $f(0)$ or y must be shown to be zero Or other valid method, e.g. sign of $f'(x)$ for $x < 0$ and $x > 0$ AG Must establish minimum clearly	
		b	f is not a one-to-one function	B1(AO2.4) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$g(0) = 0$ and $g(2) = 4$ Domain of $g^{-1}(x)$ is range of $g(x)$, so $0 \leq x \leq 4$ Range of $g^{-1}(x)$ is domain of $g(x)$, so $0 \leq g^{-1}(x) \leq 2$	B1(AO2.2a) B1(AO1.2) [2]	Must not be in terms of y Must not be an interval for x		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
31		a		B1(AO1.1) B1(AO1.1) [2]	$y = x$ straight line with positive gradient through origin $y = 2x - 1$ steeper than $y = x$ and crossing $y = x$ and y-axis as shown		
		b	DR $2x - 1 > x$ $-(2x - 1) > x$ Hen $x > 1$, $\frac{1}{3} > x$ ce $\{x : x < \frac{1}{3}\} \cup \{x : x > 1\}$	M1(AO1.1) M1(AO1.1a) A1(AO1.1) A1(AO2.5) [4]	Allow $2x - 1 = x$ Allow $-(2x - 1) = x$ For either of these inequalities Must be in this set notation form or else stated '$x < \frac{1}{3}$ or $x > 1$' as		
			Total	9			

Question		Answer/Indicative content	Marks	Part marks and guidance		
32		EITHER $f(2) = 3 \times 2^3 - 8 \times 2^2 + 3 \times 2 + 2 + 24 - 32 = 6 + 2 = 0$ Therefore by the factor theorem $(x - 2)$ is a factor OR $f(x) = (x - 2)(3x^2 - 2x - 1)$ No remainder so $(x - 2)$ is a factor	M1 (AO 1.1a) A1 (AO 1.1b) E1 (AO 2.2a) [3] M1 A1 E1 [3]	AG Function notation need not be used Zero must be seen Reason required Using algebraic division as far as $3x^2$ Correct quotient Reason required		
				Examiner's Comments There were many good answers and candidates could choose whether to use the factor theorem or to divide. Some candidates lost the mark as they presented the evidence but did not write that $f(2) = 0$ implied that $(x - 2)$ is a factor, or that no remainder implied that $(x - 2)$ is a factor. Exemplar 1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$f(x) = 2x^3 - 8x^2 + 3x + 1$ $f(2) = 2(2)^3 - 8(2)^2 + 3(2) + 1$ $= 16 - 32 + 6 + 1$ $= -9$	
					This exemplar shows why some candidates were only credited 2 of 3 marks.	
			Total	3		
33			$-5 < 2x + 1 < 5$ $-6 < 2x < 4$ $-3 < x < 2$	M1 (AO 2.1) A1 (AO 1.1) A1 (AO 1.1) [3]	$-(2x + 1) < 5$ or $2(2x + 1)^2 < 25$ $5 \text{ oe and } 2x + 1 < 5$ $-3 \text{ and } 2$ if M0 allow B1 for either condition identified $-3 < x < 2$ allow $x > -3$ and $x < 2$	
			Total	3		

Examiner's Comments

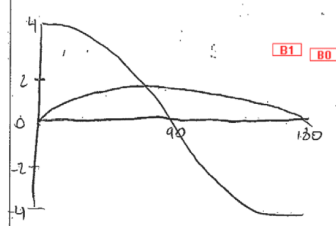
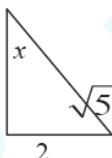
This question proved routine for the majority of candidates.

Question			Answer/Indicative content	Marks	Part marks and guidance	
34		a	$C = 2$ $A = 62$ $B = 10$	B1 (AO 3.3) B1 (AO 3.3) B1 (AO 1.1) [3]	since max when $t = 2$ since max when $(t - 2)^2 = 0$ from substitution of 22, 62 and 2 Examiner's Comments Candidates who did well in this question recognised that the maximum value of p has to be 62, and that this occurs when $t = 2$, thus obtaining A and C. The value of B soon follows. This who did less well wrote down three equations in three variables and often went astray.	
		b	substitution of 0.75 in $p = 62 - 10(t - 2)^2$ 46	M1 (AO 3.4) A1 (AO 1.1) [2]	FT <i>their</i> 2, 62, 10 allow 46.375 rounded to 2 or more sf Examiner's Comments Candidates who did well made the correct substitution in their formula. Candidates who did less well substituted $t = 45$.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$62 - 10(t - 2)^2 = 0$ $[t =]$ 4 hours 29 minutes or 4 hours 30 minutes	M1 (AO 3.4) A1 (AO 2.4) [2]	or $\geq$ or > 0 for M1 NB $t = 2 +$ $\sqrt{6.2}$ allow 4.49 or 4.5 [hours]	Examiner's Comments Candidates who did well understood that the value of p couldn't be negative and obtained a value for t accordingly. Candidates who did less well worked with $<$ instead of $=$ or $>$.
		d	substitution of $t = 1, 3$ and 5 awrt $59.4 \approx 59$ awrt $83.8 \approx 84$ awrt $88.8 \approx 89$	M1 (AO 3.4) A1 (AO 3.5a) [2]	or awrt 59.4, 83.8 and 88.8 found and supporting comment made eg they are ap proximately the same as the values in the table	if M0 allow SC1 for two values correctly found and shown to be consistent or supporting comment made

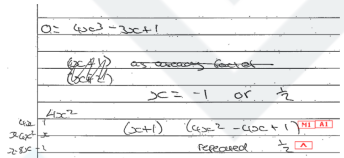
Question			Answer/Indicative content	Marks	Part marks and guidance		
		e	$p \rightarrow 90$ as $t \rightarrow$ large or when $t = 12$ $p = 89.99539 \dots$ rounded to 2 or more sf comparison with value of p for $t = 5$ eg model predicts $p = 89$ for $t = 5$ and $p = 90$ for $t = 12$ so not good advice	B1 (AO 3.5a) B1 (AO 3.5a) [2]	or model predicts $p = 90$ for (any) $t \geq 7$ so not good advice allow equivalent comment on 7 hours work for one extra mark <u>Examiner's Comments</u> Candidates who did well evaluated p at $t = 12$ and either at $t = 5$ or a value in between 5 and 12 and commented appropriately. Candidates who did less well made no supporting calculations but supplied a comment, or vice versa.		
			Total	11			

Question			Answer/Indicative content	Marks	Part marks and guidance		
35		a	$y = 4 \cos x$ $y = 2 \sin x$ 	B1 (AO1.2) B1 (AO1.2) [2]	Correct shapes and relative sizes Graphs labelled with correct amplitudes clear Ignore any graph beyond the given interval		
			Examiner's Comments The graphs were generally well done although many candidates lost a mark as they did not indicate which graph was which. Some just drew the basic graphs of $y = \sin x$ and $y = \cos x$ with an amplitude of 1. Only rarely did candidates draw the graph of $y = \cos 4x$. Some candidates who used their calculators to produce points to plot $y = 4 \cos x$ gave too few points and drew a line from (0, 4) to (180, -4)  Make sure all graphs are fully labelled. Exemplar 1				

Question			Answer/Indicative content	Marks	Part marks and guidance	
						
		b	$4\cos x = 2\sin x$ $\tan x = 2$ $x = \arctan 2$ $4\cos x = 4 \times \frac{1}{\sqrt{5}} = \frac{4}{5}\sqrt{5}$  Point of intersection is $\left(\arctan 2, \frac{4}{5}\sqrt{5}\right)$	M1 (AO3.1a) A1 (AO1.1b) M1 (AO1.1a) A1 (AO1.1b) [4]	Forming equation leading to $\tan x = k$ oe Award for 63.4° for this first A mark Do not allow for using 63.4° Attempting to find exact value for $\cos x$ or $\sin x$ Must be a coordinate in this form <u>Examiner's Comments</u> Many candidates divided by $\cos x$ to obtain an equation for $\tan x$, but there were many who got mixed up and obtained $\tan x = \frac{1}{2} \text{ or even } \tan x = 0.$ Most did not think to use Pythagoras' theorem to find the y	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					coordinate of the point of intersection.  Where a value for $\tan x$ is known, draw a simple right angled triangle and using Pythagoras' theorem to give the values for $\sin x$ or $\cos x$.	
		c	The student's argument is invalid as the period of $\tan x$ is 180° .	E1 (AO2.3) [1]	Must state or imply that the argument is invalid and give a correct reason Allow for "roots occur every 180°" or Examiner's Comments Few candidates used the periodicity of $\tan x$ to answer this question but many correct answers explained that there would be another point of intersection for $180^\circ \leq x \leq 360^\circ$.  Make sure that you clearly state whether the argument is valid or not and make sure this does not contradict your evidence.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
36		a	$f(-1) = 4 \times (-1)^3 - 3(-1) + 1$ $= -4 + 3 + 1 = 0$ Therefore $(x + 1)$ is a factor	M1 (AO2.1) A1 (AO2.2a) [2]	DR Use of $f(-1)$ must be seen. Do not allow for algebraic division. Clear conclusion must be made	Allow without conclusion if preceded by "If $f(-1) = 0$ then $(x + 1)$ will be a factor" or similar	
					Examiner's Comments This question specified a method that was to be tested, so no marks were obtained by candidates who used algebraic division here – marks for this skill were credited in part (b). Most candidates were able to evaluate $f(-1) = 0$ but this did not obtain full marks without the detailed reasoning that this meant that $(x + 1)$ was a factor of $f(x)$.		
		b	$f(x) = (x + 1)(4x^2 - 4x + 1)$ $= 0$ $= (x + 1)(2x - 1)^2 = 0$ $x = -1, \frac{1}{2}$ [repeated]	M1 (AO1.1a) A1 (AO1.1b) A1 (AO1.1b) [3]	DR Attempt to divide or to factorise by inspection with $4x^2$ correct quadratic factor seen or implied by correct linear factors	Allow full credit for $(x + 1)(4x - 2)(x - 0.5)$ No marks for solving the cubic on the calculator	
					Both roots		

Question			Answer/Indicative content	Marks	Part marks and guidance		
					seen derived from 3 correct linear factors or use of quadratic formula		
					Examiner's Comments There were many good answers seen but some candidates who correctly divided then did not state what the roots of the equation were. Some candidates used their calculators to find the roots of the equation but were not able to give the correct linear factors of $f(x)$ that were needed for full marks as in this exemplar. Exemplar 2 		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
37		a	DR $108 - 108 - 3 + 3 = 0$	B1 (AO 2.4) [1]	Sub. $x = 3$ and correct completion		
		b	DR $(x - 3)(4x^2 - 1)$ $(x - 3)(2x - 1)(2x + 1)$ $(3, 0), (\frac{1}{2}, 0), (-\frac{1}{2}, 0)$	M1 (AO 1.1a) M1 (AO 1.1) A1 (AO 1.1) A1 (AO 2.2a) [5]	For at least 2 correct x -values For all three correct points		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	DR $\int_{-\frac{1}{2}}^{\frac{1}{2}} (4x^3 - 12x^2 - x + 3) \, dx - \int_{\frac{1}{2}}^3 (4x^3 - 12x^2 - x + 3) \, dx$ $\left[x^4 - 4x^3 - \frac{1}{2}x^2 + 3x \right]_{-\frac{1}{2}}^{\frac{1}{2}} - \left[x^4 - 4x^3 - \frac{1}{2}x^2 + 3x \right]_{\frac{1}{2}}^3$ $\left(\frac{15}{16} - \frac{(-17)}{16} \right) - \left(-22\frac{1}{2} - \frac{15}{16} \right)$ $25\frac{7}{16}$	M1 (AO 3.1a) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.1) [5]	May be awarded at any stage for adding <i>their</i> two absolute areas Integration: at least one term correct All terms correct Use of <i>their</i> two pairs of limits awrt 25.4	Ignore limits (or absence of limits) for these marks	
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
38		a	Revenue R is $P \times Q$, so $R = mQ^2 + cQ$	B1 (AO 3.3) [1]			
		b	$\frac{dR}{dQ} = 2mQ + c$ $\frac{dR}{dQ} = 0 \text{ when } Q = -\frac{c}{2m}$ Maximum since $\frac{d^2R}{dQ^2} = 2m < 0 \text{ as } m \text{ is negative}$ When $Q = -\frac{c}{2m}, P = -\frac{c}{2} + c = \frac{c}{2}$ So $PED = \frac{P}{mQ} = \frac{\frac{1}{2}c}{-\frac{1}{2}c} = -1$	M1 (AO 3.1b) M1 (AO 1.1) A1 (AO 2.4) M1 (AO 1.1) E1 (AO 2.1) [5]	Or attempt to complete the square oe identification of $Q = -\frac{c}{2m}$ oe justification for maximum AG Correct working needed	$m \left[\left(Q + \frac{c}{2m} \right)^2 - \left(\frac{c}{2m} \right)^2 \right]$	
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
39		a	$f(2) = 16 - 2 = 14$. Since $f(x)$ passes through 2, 14), $f^{-1}(x)$ must pass through (14, 2)	E1 (AO2.4) [1]		
		b	$f'(x) = 4x^3 - 1$ $f'(2) = 31$ $\frac{1}{31}$	B1 (AO2.1) M1 (AO1.1) A1 (AO1.1) [3]	Accept BC	
			Total	4		
40			Either $4x - 5 = 3$ giving $x = 2$ Or $4x - 5 = -3$ $4x = 2$ giving $x = 0.5$	B1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) [3]	For answer $x = 2$ www May be implied	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
41		a	$ff(x) = f\left(\frac{2x-3}{x-2}\right) = \frac{2\left(\frac{2x-3}{x-2}\right) - 3}{\frac{2x-3}{x-2} - 2}$ $= \frac{2(2x-3) - 3(x-2)}{2x-3 - 2(x-2)}$ $= \frac{x}{1} = x$	M1 (AO 1.2) M1 (AO 1.1a) A1 (AO 2.1) [3]	Using $\frac{2x-3}{x-2}$ in the 2 formula for $f(x)$ Attempt to simplify the fractions AG Clear argument needed		
		b	$f^{-1}(x) = f(x) = \frac{2x-3}{x-2}$	B1 (AO 2.2a) [1]	Allow for statement that f is a self-inverse function		
		c	New equation is $y = f(x+1) - 1$ $y = \frac{2(x+1)-3}{(x+1)-2} - 1 = \frac{2x-1-(x-1)}{x-1}$ $y = \frac{x}{x-1}$	M1 (AO 1.2) M1 (AO 1.1a) A1 (AO 1.1) [3]	soi Attempt to combine into a single fraction		
			Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance		
42			Attempt to find Quotient Quotient is $x^2 - 2x + 2$ Remainder is -1	M1(AO 1.1a) A1(AO 1.1) B1(AO 1.1) [3]	Allow M1 for $x^2 \pm 2x$ seen in quotient and $\pm 2x^2 \pm 2x$ seen in long division.		
			Total	3			
43			-2 $5, -5$	B1(AO 1.1a) B1(AO 1.1) [2]	Both needed		
			Total	2			

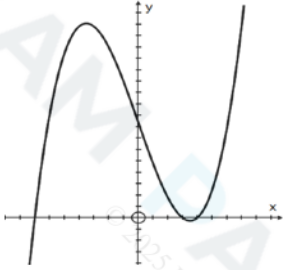
Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
44		a	5 million	B1(AO 3.4) [1]		
		b	The population is growing At a rate proportional to the population	B1(AO 3.2a) B1(AO 3.2a) [2]	or at 2% a year or by the same percentage each year	
		c	DR $5 \times 1.02^n = 10 \Rightarrow 1.02^n = 2$ $n = \frac{\log 2}{\log 1.02} = 35.002...$ 2035	M1(AO 3.4) M1(AO 1.1a) A1(AO 3.2a) [2]	oe	
		d	$\log_{10} P = \log_{10} 5 + n \log_{10} 1.02$ Of form $y = mx + c$ [with $\log_{10} P$ as y and n as x]	M1(AO 1.1) E1(AO 2.4) [2]	oe with clear link to gradient and intercept	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
45		a	$[g(x)=]3x - 6$	B1(AO 2.2a) [1]		
		b	$\{ x : x \in i, x \neq 2 \}$	B1(AO 1.1) [1]	oe; condone omission of $x \in i$	
		c	Stretch Translation Correct details for 1st transformation; can be any of (A) [stretch] parallel to x-axis scale factor $\frac{1}{3}$ or (B) [stretch] parallel to y-axis scale factor $\frac{1}{3}$ or (C) [translation] 6 to the right or (D) [translation] 2 to the right Correct details for 2nd transformation, which must correspond to the 1st transformation given: (A) [translation] 2 to the right or (B) [translation] 2 to the right or (C) [stretch] parallel to x-axis scale factor $\frac{1}{3}$ or (D) [stretch] parallel to y-axis scale factor $\frac{1}{3}$	B1(AO 1.1) B1(AO 1.1) B1(AO 1.1) B1(AO 2.2a) [4]	Allow these two marks irrespective of the order of the transformations	Transformation details not required here either Accept translation details in column-vector form or in equivalent words
			Total	6		

Question		Answer/Indicative content	Marks	Part marks and guidance		
46	a	DR Nigel should substitute $x = -7$ not $x = 7$; his calculation shows that $(x - 7)$ is not a factor $(-7)^3 - 37 \times (-7) + 84 = -343 + 259 + 84 = 0$, so $(x + 7)$ is a factor	B1 (AO2.3) B1 (AO2.1) [2]	Any suitable comment that references the factor theorem Clear argument needed	Allow for correct algebraic division if comment made about no remainder.	Examiner's Comments Many candidates understood the use of the factor theorem here but not all were able to explain sufficiently clearly to obtain full marks. Exemplar 2 <i>As he didn't sub in -7 since $x + 7 = 0$</i> $(-7)^3 - 37(-7) + 84 = 0$ $x = -7$ This exemplar is a case where the candidate did not have a very clear explanation of why Nigel's argument was wrong but was considered to be just enough. However, it is a very good example how the second mark can be lost if there is no conclusion given after finding the zero value.



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	DR $x^3 - 37x + 84 = (x + 7)(x^2 - 7x + 12)$ $= (x + 7)(x - 3)(x - 4)$ Crosses x-axis at $(-7, 0)$, $(3, 0)$, $(4, 0)$ Crosses y-axis at $(0, 84)$ 	M1 (AO3.1a) M1 (AO1.1a) A1 (AO1.1) (dep) B1 (AO1.1) B1 (AO1.1) [5]	Attempt to divide cubic by $(x + 7)$ or find a quadratic factor by inspection Attempt to factorise their quadratic factor or find points on x-axis FT Accept values shown on sketch graph www Accept value shown on sketch graph www Axes labelled and correct general shape FT their values as long as right way up.	May be seen above (Allow if at least two correct terms) The A mark is dependent on the second M mark Exact position of the stationary points is not needed	
				Examiner's Comments This is a question requiring detailed reasoning, so the method marks were awarded for finding a quadratic factor of the cubic and then obtaining the linear factors leading to the points where the graph crosses the x-axis. Examiners needed to be		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					sure that the correct graph had not been obtained by simply typing the function into a graphical calculator and copying whatever they saw. The B marks for the y intercept and the general shape of the graph would have been still available but many candidates lost the final mark as they did not label their axes. (In order not to penalise the same mistake twice, this condition was not applied a second time in Q 10(a).)	
	c		DR Equation of the form $y = (x - 1)^3 - 37(x - 1) + 84$ $= (x^3 - 3x^2 + 3x - 1) - 37(x - 1) + 84$ $y = x^3 - 3x^2 - 34x + 120$ Alternative solution Equation of the form $y = (x - 1 + 7)(x - 1 - 3)(x - 1 - 4)$ $= (x + 6)(x^2 - 9x + 20)$ or $(x - 4)(x^2 + x - 30)$ or $(x - 5)(x^2 + 2x - 24)$ $y = x^3 - 3x^2 - 34x + 120$	M1 (AO3.1a) M1 (AO1.1a) A1 (AO1.1) A1 (AO1.1) [4] M1 M1 A1 A1 [4]	Substituting $(x - 1)$ for x twice Attempt to expand Correct expansion of their cubic term cao (including y =) Substituting $(x - 1)$ for x everywhere Attempt to multiply out their factors Correct quadratic factor cao (including y =)	Allow for $(x + 6)(x - 4)(x - 5)$
					Examiner's Comments Many candidates understood the change to	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					the algebra needed for a sideways translation but many candidates had their translation in the opposite direction by using $f(x + 1)$ instead. Most had efficient ways to expand the brackets to obtain the correct expansion but the omission of "$y =$" cost many candidates the final mark.  AfL Many candidates did not include $y =$ in their answer despite it being included in the question and lost the final mark. In preparation for examinations, teachers need to stress the importance of accurate use of notation, and the value of checking how an answer needs to be given.	
			Total	11		

Question		Answer/Indicative content	Marks	Part marks and guidance		
47	a	$(2x + 3 - 1)^2$ or $(2x + 3)^2 - 2(2x + 3) + 1$ seen simplified to eg $4(x + 1)^2$ or $4x^2 + 8x + 4$ or $(2x + 2)^2$ domain is $-1 < x < 0$	M1 (AO1.1) A1 (AO1.1) B1 (AO1.1) [3]	substitution mark the final answer from $2x + 3 > 1$	ignore superfluous work on eg finding roots	
				Examiner's Comments Candidates who did well in this question were able to complete the algebraic manipulation successfully and understood how to find the correct domain for the composite function. Candidates who did less well did not deal with the domain successfully and / or spoiled their answer for the formula by dividing unnecessarily by 4.		
	b	$0 < gf(x) < 4$	B1 (AO1.1) [1]	Examiner's Comments Very few candidates were able to find this successfully and generally this topic seems to be poorly understood.		

Question		Answer/Indicative content	Marks	Part marks and guidance	
	c	factorise their $gf(x)$ to obtain perfect square or complete the square $y = 4(x + 1)^2$ or $(2x + 2)^2$ oe $(x + 1) = (\pm)\sqrt{\frac{y}{4}}$ oe $[(gf)^{-1}(x) =] \sqrt{\frac{x}{4}} - 1$ or $\frac{\sqrt{x}}{2} - 1$ oe domain is $0 < x < 4$	M1 (AO3.1a) A1 (AO2.1) M1 (AO1.1) A1 (AO1.1) B1 (AO1.1) [5]	allow eg $2(x + 1)(2x + 2)$; may follow slip eg dividing by 4 FT FT their (b)	or $g^{-1}(x) = \sqrt{x} + 1$ or $f^{-1}(x) = \frac{1}{2}(x - 3)$ for M1 A1 for both correct M1 for their f^{-1} (the $\sqrt{x} + 1$) ir $(gf)^{-1}(x) = \sqrt{\frac{x}{4}} - 1$ A1 for or $\frac{\sqrt{x}}{2} - 1$ oe x and y may be interchanged for the first 3 marks but not for the final A1
				Examiner's Comments Candidates who did well in this question understood how to "make y the subject" either by completing the square or expressing their answer to part (a) as a perfect square. They wrote their answer in the form "$(gf)^{-1} =$" as opposed to "$y =$". The very best candidates made the connection between their	

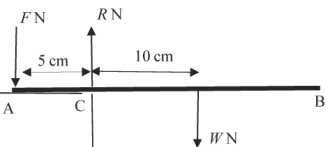
Question			Answer/Indicative content	Marks	Part marks and guidance	
					answer to above part and the domain for the inverse of the composite function. Candidates who did less well left their answer in the form "$y =$" and made no attempt to identify a domain. Candidates who did not do well subtracted $8x$ from each side and then took square roots.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance		
48	a	i	$[\text{Area}] = 2xl + \pi x^2$ $l = \pi(r - x) \text{ so}$ $\text{Area} =$ $2x\pi(r - x) + \pi x^2 = 2\pi rx - \pi x^2$ $= \pi x(2r - x)$	M1 (AO1.1a) A1 (AO2.4) [2]	Use of $l = \pi(r - x)$ to get formula in terms of x Convincing completion (answer given) Examiner's Comments Part (i) saw many correct answers, although some messy presentation made it difficult to follow at times. With the expression given there was the tendency to see some set up an incorrect equation and then try to fudge the answer.		
		ii	$ry + hx = hr \text{ so}$ $x = \frac{hr - ry}{h} \text{ and}$ $\pi x(2r - x) = \pi \left(\frac{hr - ry}{h} \right) \left(\frac{2rh - hr + ry}{h} \right)$ $\pi \left(\frac{hr - ry}{h} \right) \left(\frac{hr + ry}{h} \right) = \frac{\pi}{h^2} (h^2 r^2 - r^2 y^2)$ $= \frac{\pi r^2}{h^2} (h^2 - y^2)$	M1 (AO3.1a) A1 (AO2.2a) [2]	Finding x in terms of y correctly and using it Convincing completion (answer given) Examiner's Comments Part (ii) saw a lot of correct attempts, but few seem to see the simplest route through the algebra. Some tend to make their expressions more complicated than is necessary, with there then being more room for errors, as with those who are messy in their presentation.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
	b	i	$y = 0$ so area = $\frac{\pi r^2}{h^2}(h^2) = \pi r^2$	B1 (AO3.5a) [1]	<u>Examiner's Comments</u> For those who attempted (i), the most common error seemed to be substituting $y = 0$ and $h = 1$.	
		ii	$y = h$ so area = $\frac{\pi r^2}{h^2}(h^2 - h^2) = 0$	B1 (AO3.5a) [1]	<u>Examiner's Comments</u> Part (ii) was similar to above, with candidates not realising that all that was required was to use $y = h$.	
			Total	6		

Question	Answer/Indicative content	Marks	Part marks and guidance
49	$r = \frac{5.4}{\pi}$ V = 74.2 to 74.3 or $r = 1.727$ or 1.73 Suitable comment consistent with working E.g. Not exactly right but the measurements may not be exact The volume / radius is too small but it's close Allowing for approximation the formula seems accurate	B1 (AO3.1a) B1 (AO1.1) E1 (AO3.5b) [3]	soi by 1.71[88], 1.72 If r from V = 75 and h = 12 Dependent on both B marks <u>Examiner's Comments</u> Very often done well, with a simple minimal comment about being accurate sufficing for full marks. The most common approach was to find the volume $r = \frac{5.4}{\pi}$ when π, although some did find the value of r from volume=75 and proceeded to correctly compare with the decimal version of $\frac{5.4}{\pi}$. The most common error seen was the use of $r = \frac{5.4}{2}$. <u>Exemplar 9</u> $V = \frac{4}{3} \times \pi \times \left(\frac{5.4 \times 10^{-2}}{\pi} \right)^2 \times 12 \times 10^{-2}$ $= 7.42 \times 10^{-5} \text{ m}^3$ $\approx 7.4 \times 10^{-5} \text{ m}^3$ the model predicted $7.4 \times 10^{-5} \text{ m}^3$ when we needed $7.5 \times 10^{-5} \text{ m}^3$ and therefore was less than 1% out. That is a significant level that allows me to say the model is sufficiently accurate. However, because the model underpredicts the accuracy needs to be changed because mathematically it's got to work in the real world would need to overestimate. I included this example as

Question			Answer/Indicative content	Marks	Part marks and guidance	
					the candidate had only scored 1 mark so far in the comprehension and given 'No response' to half of the 8 previous questions. But notice that, with presumably a very sketchy understanding of the whole situation, the candidate could still score full marks on this part – the final question on the paper. The comprehension is definitely an area where resilience is to be encouraged however difficult previous questions have been.	
			Total	3		

Question		Answer/Indicative content	Marks	Part marks and guidance		
50	a		B1(AO1.1a) B1(AO1.1) [2]	F and their weight even if not labelled in roughly correct position Distances labelled. Allow for 5cm and any measurements that show that the weight is acting in the centre of the ruler.	R is already drawn in Printed Answer Booklet	
	b	Take moments about C $10 \times 0.028g = 5F$ $F = 0.549 \text{ N}$	M1(AO1.1a) A1(AO1.1) [2]	Moments about any point with all relevant forces seen. Allow their weight and one incorrect distance allow $\frac{7}{125}g \text{ or}$	[Value of R is 0.8232 N]	
		Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
51	a		DR $f(-2) = (-2)^3 + (-2)^2$ $-8(-2)-12 = 0$ so [by the factor theorem] $(x + 2)$ is a factor	M1(AO1. 1a) A1(AO2. 2a) [2]	Substitution seen. Do not allow for division here Clear conclusion.		
	b		DR $f(x) = (x + 2)(x^2 - x - 6)$ $f(x) = (x + 2)^2 (x - 3) = 0$ so $x = 3$ or $x = -2$ [repeated]	M1(AO3. 1a) A1(AO1. 1) B1(1.1) A1(AO2. 1) [4]	Attempt to divide or factorise Correct quadratic factor seen Product of linear factors seen Do not allow without full working	Also allow M1 A1 for $f(x) = (x - 3)(x^2 + 4x + 4)$ if $(x - 3)$ also established as a factor by division or factor theorem	
			Total	6			

Question		Answer/Indicative content	Marks	Part marks and guidance		
52	a	weight $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix}$ Equilibrium equation $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \mathbf{F}_2 = \mathbf{0}$ $\mathbf{F}_2 = \begin{pmatrix} -4 \\ 2+1.5g \end{pmatrix} = \begin{pmatrix} -4 \\ 16.7 \end{pmatrix} \text{ N}$	B1(AO1.2) M1(AO1.1a) A1(AO1.1) [3]	Allow seen or implied by correct answer FT their weight. Allow sign errors must be vector allow i-j form		
	b	Newton's second law $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 20 \end{pmatrix} = m\mathbf{a}$ $\mathbf{a} = \frac{1}{1.5} \begin{pmatrix} 6 \\ 3.3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2.2 \end{pmatrix} \text{ m s}^{-2}$	M1(AO1.1a) A1(AO1.1) [2]	Addition of vectors. Allow if weight missing but other two forces and acceleration seen in equation. Must be vector; any correct form.	Allow wrong weight only if given as a vector	
		Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
53	a		Acceleration due to gravity is not a constant but depends on location in the universe.	B1(AO3.5b) [1]	Allow any sensible comment that g might be different on the moon		
	b		Using $s = 2$, $u = 0$, $t^{s=ut+\frac{1}{2}at^2}$ = 1.6 and Downwards the positive direction $2 = \frac{1}{2} \times 1.6^2 a$ giving [$\alpha = 1.5625$] which 1.6 m s ⁻² to 2 sf	M1(AO3.3) A1(AO1.1) [2]	Allow any sign convention Allow -1.6 m s ⁻² if upwards is clearly indicated as positive. Must be 2 significant figures.		
	c		Using $u = 15$, $v = 0$, $a = -1.6$ and $v^2 = u^2 + 2as$ Upwards the positive direction $0 = 15^2 - 2 \times 1.6s$ $s = 70.3$ m	M1(AO3.4) A1(AO1.1) [2]	Use of $suvat$ equation(s) leading to a value for s . Allow sign errors Must follow from correct working and if negative, explained.	Allow answers in the range 70 to 72.2	
			Total	12			